

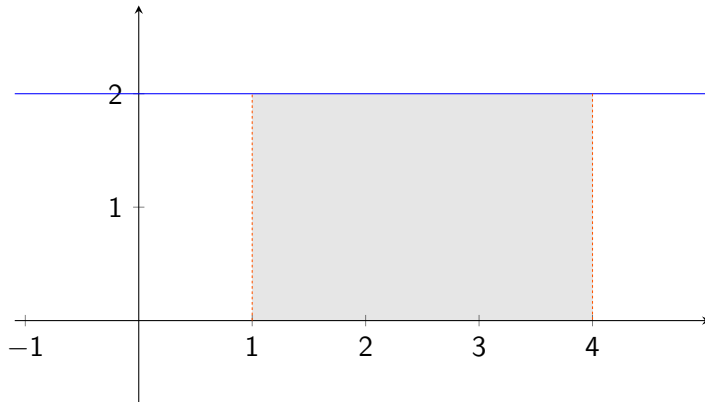
# TCX2101: Calculus and Linear Algebra for Computing

## Chapter 4: Integration and Applications

# Definite Integral

## Area of Rectangle

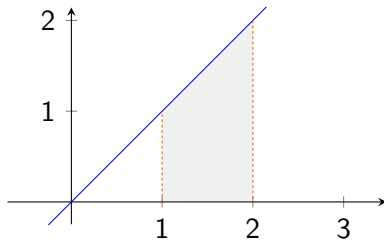
What is the area of the shaded region?



$$\text{Area} = (4 - 1) \times 2 = 6.$$

# Area of Trapezium

What is the area of the shaded region?



Using the formula for area of a trapezium, the area is  $\frac{1+2}{2} = \frac{3}{2}$ .

Can we use rectangles to estimate the area?

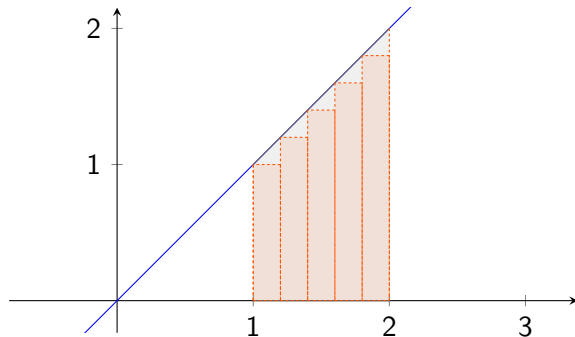
## Area of Trapezium: Lower Sum

To approximate the area under the curve using rectangles, we first split the base interval  $[1, 2]$  into five equal subintervals, each with width

$$\Delta x = \frac{2 - 1}{5} = 0.2.$$

In each subinterval, construct a rectangle whose height is the **minimum value** of  $f(x)$  within that subinterval.

This ensures that the entire rectangle **lies completely under** the graph, while still touching it.



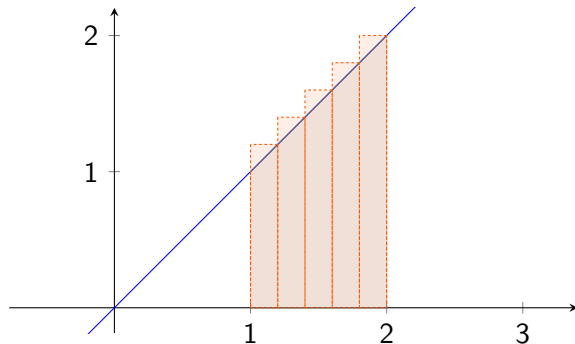
$$\text{Area} = 0.2f(1) + 0.2f(1.2) + 0.2f(1.4) + 0.2f(1.6) + 0.2f(1.8) = 1.4$$

Since each rectangle is deliberately chosen to sit entirely below the curve, this approximation is **strictly less than** the true area.

## Area of Trapezium: Upper Sum

Alternatively, in each subinterval, construct a rectangle whose height is the **maximum value** of  $f(x)$  within that subinterval.

This ensures that the entire rectangle **lies completely above** the graph, while still touching it.



$$\text{Area} = 0.2f(1.2) + 0.2f(1.4) + 0.2f(1.6) + 0.2f(1.8) + 0.2f(2) = 1.6$$

Since each rectangle is deliberately chosen to sit entirely above the curve, this approximation is **strictly more than** the true area.

# Riemann Sum

What happens if we use more, thinner rectangles?

Split the interval  $[1, 2]$  into  $n$  equal subintervals. Then each subinterval has width

$$\Delta x = \frac{2 - 1}{n} = \frac{1}{n}.$$

Since  $f(x)$  is increasing,

- ▶ the lower sum  $L_n$  uses the *left endpoints* of each subinterval,

$$L_n = \sum_{k=0}^{n-1} f\left(1 + k\frac{1}{n}\right) \Delta x = \sum_{k=0}^{n-1} \left(1 + k\frac{1}{n}\right) \cdot \frac{1}{n} = \frac{3}{2} - \frac{1}{2n};$$

- ▶ the upper sum  $U_n$  uses the *right endpoints* of each subinterval,

$$U_n = \sum_{k=1}^n f\left(1 + k\frac{1}{n}\right) \Delta x = \sum_{k=1}^n \left(1 + k\frac{1}{n}\right) \cdot \frac{1}{n} = \frac{3}{2} + \frac{1}{2n}.$$

# Riemann Sum

As we increase the number of subintervals  $n \rightarrow \infty$  (so  $\Delta x \rightarrow 0$ ),

$$\lim_{n \rightarrow \infty} L_n = \lim_{n \rightarrow \infty} U_n = \frac{3}{2},$$

which is the true area. In other words, the upper and lower sums *squeeze* closer and closer together, and both approach the same limiting value  $\frac{3}{2}$ , the exact area under the curve from  $x = 1$  to  $x = 2$ .

<https://www.geogebra.org/m/qe6jMAK2>



## Definite Integral

Let  $f(x)$  be a function defined on the interval  $[a, b]$ . Subdivide the interval into  $n$  equal subintervals, each of width

$$\Delta x = \frac{b-a}{n}, \quad x_k = a + k\Delta x, \quad k = 0, \dots, n.$$

For each  $n$ :

- ▶ The lower sum  $L_n$  is formed by taking, in each subinterval, the **minimum value** of  $f(x)$ :

$$L_n = \sum_{k=1}^n \left( \min_{x \in [x_{k-1}, x_k]} f(x) \right) \Delta x,$$

- ▶ The upper sum  $U_n$  is formed by taking, in each subinterval, the **maximum value** of  $f(x)$ :

$$U_n = \sum_{k=1}^n \left( \max_{x \in [x_{k-1}, x_k]} f(x) \right) \Delta x.$$

Since every minimum is at most every maximum,

$$L_n \leq U_n \quad \text{for all } n.$$

# Definite Integral

If, as  $n \rightarrow \infty$ , the lower and upper sums *converge to the same limit*,

$$\lim_{n \rightarrow \infty} L_n = \lim_{n \rightarrow \infty} U_n = A, \quad \text{or} \quad \lim_{n \rightarrow \infty} (U_n - L_n) = 0$$

then we say  $f$  is integrable on  $[a, b]$ , and define the definite integral of  $f$  to be

$$\int_a^b f(x) dx = \lim_{n \rightarrow \infty} L_n = \lim_{n \rightarrow \infty} U_n.$$

# Explaining the Notation

$$\int_a^b f(x) dx$$

- (i)  $\int$  is the integral sign.
- (ii)  $a$  is the **lower limit** of the integral.
- (iii)  $b$  is the **upper limit** of the integral.
- (iv) The function  $f(x)$  is called the **integrand**.
- (v)  $x$  is a dummy variable, it is the **variable** of the integration. That is, the following definite integrals are equal,

$$\int_a^b f(x) dx = \int_a^b f(t) dt = \int_a^b f(s) ds.$$

$dx$  is commonly known as the **infinitesimal** subinterval width.

# Integrability of Continuous Functions

## Theorem

If a function  $f$  is continuous over the interval  $[a, b]$ , or if  $f$  has at most finitely many jump or removable discontinuity there, then  $f$  is *integrable* there and the definite integral

$$\int_a^b f(x) \, dx$$

exists.

Sketch of Proof.

$$\lim_{n \rightarrow \infty} (U_n - L_n) = \lim_{n \rightarrow \infty} \sum_{k=1}^n \left( \max_{x \in [x_{k-1}, x_k]} f(x) - \min_{x \in [x_{k-1}, x_k]} f(x) \right) \Delta x$$

# Integrability of Continuous Function

Proof Continue.

Since  $f(x)$  is continuous, as the interval width becomes smaller,  $x_k - x_{k-1} \rightarrow 0$ , the difference between the maximum and minimum within the interval also vanishes,

$$\left( \max_{x \in [x_{k-1}, x_k]} f(x) - \min_{x \in [x_{k-1}, x_k]} f(x) \right) \rightarrow 0 \quad \text{as} \quad x_k - x_{k-1} \rightarrow 0.$$

Hence,  $f(x)$  is integrable, and

$$\lim_{n \rightarrow \infty} (U_n - L_n) = \lim_{n \rightarrow \infty} \sum_{k=1}^n \left( \max_{x \in [x_{k-1}, x_k]} f(x) - \min_{x \in [x_{k-1}, x_k]} f(x) \right) \Delta x = 0$$

and thus the function is integrable,

$$\int_a^b f(x) dx = \lim_{n \rightarrow \infty} U_n = \lim_{n \rightarrow \infty} L_n.$$



# Properties of Definite Integral: Order of Integration

Let  $f(x)$  be a function integrable over  $[a, b]$ .

Order of integration : 
$$\int_a^b f(x) dx = - \int_b^a f(x) dx.$$

In defining  $\int_a^b f(x) dx$  as the limit  $\lim_{n \rightarrow \infty} \sum_{k=1}^n f(c_k) \Delta x$ , we moved from left to right,  $\Delta x = x_k - x_{k-1}$ .

If instead we move from right to left,  $x_0 = b$  and  $x_n = a$ , then  $\Delta x = x_{k-1} - x_k$ , and thus

$$\int_b^a f(x) dx = \lim_{n \rightarrow \infty} \sum_{k=1}^n f(c_k)(x_{k-1} - x_k) = - \lim_{n \rightarrow \infty} \sum_{k=1}^n f(c_k)(x_k - x_{k-1}) = - \int_a^b f(x) dx,$$

where  $f(c_k)$  is either  $\min_{x \in [x_{k-1}, x_k]} f(x)$  or  $\max_{x \in [x_{k-1}, x_k]} f(x)$ .

## Properties of Definite Integral: Zero Width Interval

Let  $f(x)$  be a function integrable over  $[a, b]$ .

$$\text{Zero width interval : } \int_a^a f(x) \, dx = 0.$$

For any division of the interval, have  $a = x_0 = x_1 = \cdots = x_n = a$ , then

$$\Delta x = x_{k+1} - x_k = 0 \quad \text{for all } k = 0, \dots, n-1.$$

Hence,

$$\int_a^b f(x) \, dx = \lim_{n \rightarrow \infty} \sum_{k=1}^n f(c_k) \cdot 0 = \lim_{n \rightarrow \infty} 0 = 0.$$

# Properties of Definite Integral: Scalar Multiple

Let  $f(x)$  be a function integrable over  $[a, b]$ .

$$\text{Scalar multiple : } \int_a^b kf(x) \, dx = k \int_a^b f(x) \, dx$$

This follows from the scale multiple rule for limits:

$$\begin{aligned} \int_a^b kf(x) \, dx &= \lim_{n \rightarrow \infty} \sum_{k=1}^n k \cdot f(c_k) \Delta x = k \cdot \lim_{n \rightarrow \infty} \sum_{k=1}^n f(c_k) \Delta x \\ &= k \int_a^b f(x) \, dx \end{aligned}$$



# Properties of Definite Integral: Sum and Difference

Let  $f(x)$  and  $g(x)$  be a function integrable over  $[a, b]$ .

Sum and difference : 
$$\int_a^b (f(x) \pm g(x)) \, dx = \int_a^b f(x) \, dx \pm \int_a^b g(x) \, dx$$

This follows from the sum and difference rule for limits:

$$\begin{aligned} \int_a^b (f(x) \pm g(x)) \, dx &= \lim_{n \rightarrow \infty} \sum_{k=1}^n (f(c_k) \pm g(c_k)) \Delta x = \lim_{n \rightarrow \infty} \left( \sum_{k=1}^n f(c_k) \Delta x \pm \sum_{k=1}^n g(c_k) \Delta x \right) \\ &= \lim_{n \rightarrow \infty} \sum_{k=1}^n f(c_k) \Delta x \pm \lim_{n \rightarrow \infty} \sum_{k=1}^n g(c_k) \Delta x \\ &= \int_a^b f(x) \, dx \pm \int_a^b g(x) \, dx \end{aligned}$$

# Properties of Definite Integral: Max-Min Inequality

Let  $f(x)$  be a function integrable over  $[a, b]$ .

Max-min inequality :  $\min_{x \in [a, b]} f(x)(b - a) \leq \int_a^b f(x) dx \leq \max_{x \in [a, b]} f(x)(b - a)$

$$\begin{aligned} \min_{x \in [a, b]} f(x)(b - a) &= \min_{x \in [a, b]} f(x) \cdot \sum_{k=1}^n \Delta x = \sum_{k=1}^n \min_{x \in [a, b]} f(x) \Delta x \\ &\leq \sum_{k=1}^n \min_{x \in [x_{k-1}, x_k]} f(x) \Delta x \\ &\leq \int_a^b f(x) dx. \end{aligned}$$

The proof for  $\int_a^b f(x) dx \leq \max_{x \in [a, b]} f(x)(b - a)$  is analogous.

## Properties of Definite Integral: Domination

Let  $f(x)$  and  $g(x)$  be a function integrable over  $[a, b]$  such that  $f(x) \leq g(x)$  on  $[a, b]$ .

$$\text{Domination : } \int_a^b f(x) \, dx \leq \int_a^b g(x) \, dx$$

In particular, if  $f(x) \geq 0$  on  $[a, b]$ , then  $\int_a^b f(x) \, dx \geq 0$ .

Proof is left as an exercise.

## Properties of Definite Integral: Subinterval

Let  $f(x)$  be a function integrable over the relevant intervals.

$$\text{Subinterval : } \int_a^b f(x) dx = \int_a^c f(x) dx + \int_c^b f(x) dx.$$

We sketch the proof only for the case  $c \in [a, b]$ . Subdivide the interval  $[a, b]$  so that the point  $c$  appears as an interior point of the partition:

$$x_0 = a < x_1 < \cdots < x_n = c < x_{n+1} < \cdots < x_{2n} = b.$$

$$\begin{aligned} \int_a^c f(x) dx + \int_c^b f(x) dx &= \lim_{n \rightarrow \infty} \sum_{k=1}^n f(c_k) \Delta x + \lim_{n \rightarrow \infty} \sum_{k=n+1}^{2n} f(c_k) \Delta x = \lim_{n \rightarrow \infty} \sum_{k=1}^{2n} f(c_k) \Delta x \\ &= \int_a^b f(x) dx \end{aligned}$$

## Question

In the subinterval property, do we need  $c \in [a, b]$ ? Will the property still hold if, for example,  $a < b < c$ , or when  $c < a < b$ ?

## Fundamental Theorem of Calculus and Indefinite Integral

# Introduction

Consider the function  $f(x) = x$ . Define a new function

$$F(x) = \int_0^x f(t) dt.$$

For each  $x \geq 0$ , this integral represents the area under the straight line  $f(t) = t$  from 0 to  $x$ , which is the area of a right triangle with base  $x$  and height  $x$ . Thus,

$$F(x) = \int_0^x f(t) dt = \frac{1}{2}x^2.$$

Now differentiate  $F(x)$  with respect to  $x$ :

$$\frac{d}{dx}F(x) = \frac{d}{dx}\left(\frac{x^2}{2}\right) = x = f(x).$$

Hence,

$$\frac{d}{dx} \int_0^x f(t) dt = f(x).$$

## Introduction

On the other hand, for any  $a, b \in \mathbb{R}$  with  $a < b$ , the definite integral

$$\int_a^b f(x) dx$$

represents the area under the straight line  $f(x) = x$  from  $x = a$  to  $x = b$ . This region is a trapezium whose parallel sides have lengths  $f(a) = a$  and  $f(b) = b$ , and whose base has length  $b - a$ . Therefore, its area is

$$\int_a^b f(x) dx = \frac{1}{2}(b - a)(a + b) = \frac{b^2}{2} - \frac{a^2}{2}.$$

Since  $F(x) = \frac{x^2}{2}$ , we obtain

$$\int_a^b f(x) dx = F(b) - F(a).$$

The above results are known as the **fundamental theorem of calculus**.



# Fundamental Theorem of Calculus I

## Theorem (Fundamental Theorem of Calculus I)

If  $f$  is *continuous* on  $[a, b]$ , then

$$F(x) = \int_a^x f(t) dt$$

is *continuous* on  $[a, b]$  and *differentiable* on  $(a, b)$  with derivative  $f(x)$ ,

$$F'(x) = \frac{d}{dx} \int_a^x f(t) dt = f(x).$$

Proof.

$$F'(x) = \lim_{h \rightarrow 0} \frac{F(x+h) - F(x)}{h} = \lim_{h \rightarrow 0} \frac{1}{h} \left( \int_a^{x+h} f(t) dt - \int_a^x f(t) dt \right) = \lim_{h \rightarrow 0} \frac{1}{h} \int_x^{x+h} f(t) dt.$$

# Fundamental Theorem of Calculus I

Proof Continue.

For small  $h$ , the interval  $[x, x + h]$  is very short, and since  $f$  is continuous,

$$\int_x^{x+h} f(t) dt \approx f(x) \cdot ((x+h) - x) = f(x) h.$$

Hence,

$$\frac{1}{h} \int_x^{x+h} f(t) dt \approx \frac{1}{h} (f(x) h) = f(x),$$

and letting  $h \rightarrow 0$  gives

$$F'(x) = f(x).$$



## Example

1. Find  $\frac{d}{dx} \int_0^x \sqrt{1-t^2} dt$ .

$$\frac{d}{dx} \int_0^x \sqrt{1-t^2} dt = \sqrt{1-x^2}.$$

2. Find  $\frac{d}{dx} \int_x^\pi t^2 \sin(t) dt$ .

$$\frac{d}{dx} \int_x^\pi t^2 \sin(t) dt = -\frac{d}{dx} \int_\pi^x t^2 \sin(t) dt = -x^2 \sin(x).$$

3. Find  $\frac{d}{dx} \int_0^{x^2} e^t dt$ .

Let  $u(x) = x^2$ , then by chain rule,

$$\frac{d}{dx} \int_0^{u(x)} e^t dt = \frac{d}{du} \int_0^u e^t dt \cdot \frac{d}{dx} u(x) = 2xe^{x^2}.$$

## Question

Recall that if  $f'(x) = g'(x)$  on the interval  $(a, b)$ , what can we conclude about  $f(x)$  and  $g(x)$ ?

## Question

Recall that if  $f'(x) = g'(x)$  on the interval  $(a, b)$ , what can we conclude about  $f(x)$  and  $g(x)$ ?

By the corollary of the Mean Value Theorem, we can conclude that

$$f(x) = g(x) + C$$

for some constant  $C$  on  $(a, b)$ .

# Antiderivative

## Definition

A function  $F$  is an antiderivative of  $f$  on an interval  $I$  if

$$F'(x) = f(x)$$

for all  $x \in I$ .

## Remarks

1. Antiderivatives are **not unique**. For example,

$$\frac{d}{dx}x^2 = \frac{d}{dx}(x^2 + 3) = 2x.$$

That is, both  $x^2 + 3$  and  $x^2$  are antiderivatives of  $2x$ .

2. By the result recalled in the previous slide, we conclude that **any two antiderivatives differ by a constant**. That is, if  $F$  and  $G$  are antiderivatives of  $f$  on  $I$ , then

$$G(x) = F(x) + C.$$

3. By the fundamental theorem of calculus I,

$$F(x) = \int_a^x f(t) dt$$

is an **antiderivative** of  $f(x)$ .

# Fundamental Theorem of Calculus II

## Theorem (Fundamental Theorem of Calculus II)

If  $f$  is *continuous* over  $[a, b]$  and  $F$  is any *antiderivative* of  $f$  on  $[a, b]$ , then

$$\int_a^b f(x) \, dx = F(b) - F(a).$$

We will use the notation

$$[F(x)]_a^b := F(b) - F(a).$$



# Fundamental Theorem of Calculus II

Proof.

By the Fundamental Theorem of calculus I,

$$G(x) = \int_a^x f(t) dt$$

is an antiderivative of  $f(x)$ . So, if  $F(x)$  is any antiderivative of  $f(x)$ , then  $F(x) = G(x) + C$  for some constant  $C$ . Evaluating

$$\begin{aligned} F(b) - F(a) &= (G(b) + C) - (G(a) + C) = G(b) - G(a) \\ &= \int_a^b f(t) dt - \int_a^a f(t) dt = \int_a^b f(t) dt - 0 \\ &= \int_a^b f(t) dt \end{aligned}$$



The proof also demonstrates that the evaluation is independent of the choice of antiderivative.

## Example

1. Since  $\frac{d}{dx} \sec(x) = \sec(x) \tan(x)$ ,

$$\int_{-\frac{\pi}{4}}^0 \sec(x) \tan(x) dx = [\sec(x)]_{-\frac{\pi}{4}}^0 = \sec(0) - \sec\left(-\frac{\pi}{4}\right) = 1 - \sqrt{2}.$$

2. Since  $\frac{d}{dx} \ln(x) = \frac{1}{x}$ ,

$$\int_1^2 \frac{1}{x} dx = [\ln(x)]_1^2 = \ln(2) - \ln(1) = \ln(2).$$

- 3.

$$\int_0^1 \frac{1}{x^2 + 1} dx = [\tan^{-1}(x)]_0^1 = \tan^{-1}(1) - \tan^{-1}(0) = \frac{\pi}{4}.$$

# Definition of Indefinite Integral

## Definition

The collection of all antiderivatives of  $f$  is called the **indefinite integral** of  $f$ , and is denoted by

$$\int f(x) \, dx.$$

For any antiderivative  $F$  of  $f$ , let

$$F(x) + C$$

denote the set of all antiderivatives of  $F$ .

Thus we may denote the set of all antiderivatives of  $f$  by

$$\int f(x) \, dx = F(x) + C$$

for some antiderivative  $F$  of  $f$ .

## Remarks

- (i) The expression  $F(x) + C$  is a *symbol* representing the entire family of antiderivatives of  $f$ ; here  $C$  is not a particular number, but a placeholder indicating that *any* constant may be chosen. One may think of  $C$  as a variable that can take any real value. Each choice of  $C \in \mathbb{R}$  gives a different member of the antiderivative family.

## Remarks

- (ii) The **differential operator**  $\frac{d}{dx}$  takes a function and returns a single function. The **indefinite integral operator**  $\int dx$  takes a function and returns a set of functions (the set of all antiderivatives on an interval). The **definite integral**  $\int_a^b dx$  takes a function and returns a real number.

$$\begin{array}{ccc} f(x) & \xrightarrow{\frac{d}{dx}} & f'(x) \\ \{ F(x) + C \mid C \in \mathbb{R} \} & \xleftarrow{\int dx} & f(x) \\ \int_a^b f(x) dx = F(b) - F(a) & \xleftarrow{\int_a^b dx} & f(x) \end{array}$$

## Remarks

(iii) If an **initial condition** is given, then we solve for the constant  $C$ . For example, suppose

$$F(x) = \int 3x^2 dx, \quad F(1) = -1.$$

Then

$$F(x) = x^3 + C.$$

The initial condition gives

$$-1 = F(1) = 1 + C \Rightarrow C = -2.$$

Thus,

$$F(x) = x^3 - 2.$$

## Examples

$$1. \int x^r dx = \frac{1}{r+1} x^{r+1} + C, \quad r \in \mathbb{Q} \setminus \{-1\}$$

$$2. \int \sin(kx) dx = -\frac{1}{k} \cos(kx) + C$$

$$3. \int \cos(kx) dx = \frac{1}{k} \sin(kx) + C$$

$$4. \int \sec^2(kx) dx = \frac{1}{k} \tan(kx) + C$$

$$5. \int \csc^2(kx) dx = -\frac{1}{k} \cot(kx) + C$$

$$6. \int \sec(kx) \tan(kx) dx = \frac{1}{k} \sec(kx) + C$$

$$7. \int \csc(kx) \cot(kx) dx = -\frac{1}{k} \csc(kx) + C$$

$$8. \int e^{kx} dx = \frac{1}{k} e^{kx} + C$$

$$9. \int \frac{1}{x} dx = \ln(x) + C, \quad x > 0$$

$$10. \int \frac{1}{x} dx = \ln(|x|) + C, \quad x < 0$$

$$11. \int \frac{1}{\sqrt{1-(kx)^2}} dx = \frac{1}{k} \sin^{-1}(kx) + C, \quad x < \frac{1}{k}$$

$$12. \int \frac{1}{1+(kx)^2} dx = \frac{1}{k} \tan^{-1}(kx) + C$$

$$13. \int \frac{1}{x\sqrt{(kx)^2-1}} dx = \sec^{-1}(kx) + C, \quad x > \frac{1}{k}$$

$$14. \int a^{kx} dx = \frac{a^{kx}}{k \ln(a)} + C, \quad a > 0$$

**Exercise:** Prove the above equations. Hint: Differentiate the RHS.

## Examples

1. Evaluate  $\int (2x + 1)^2 dx$ .

Since  $\frac{d}{dx}(2x + 1)^3 = 3(2x + 1)^2(2) = 6(2x + 1)^2$ , by linearity of differentiation,  $\frac{d}{dx} \frac{(2x+1)^3}{6} = (2x + 1)^2$ .  
Hence,

$$\int (2x + 1)^2 dx = \frac{(2x + 1)^3}{6} + C.$$

2. Evaluate  $\int (x^2 - 3)^2 dx$ .

$$\int (x^2 - 3)^2 dx = \int x^4 - 6x^2 + 9 dx = \frac{x^5}{5} - \frac{6x^3}{3} + 9x + C = \frac{x^5}{5} - 2x^3 + 9x + C.$$

3. Evaluate  $\int \frac{\sqrt{t} + 1}{t^2} dt$ .

$$\int \frac{\sqrt{t} + 1}{t^2} dt = \int t^{-\frac{3}{2}} + t^{-2} dt = -\frac{2}{\sqrt{t}} - \frac{1}{t} + C = C - \frac{2\sqrt{t} + 1}{t}.$$



## Area Under and Between Graphs

# Nonnegative and Nonpositive Functions

- ▶ We have seen that if  $f(x) \geq 0$  and integrable over  $[a, b]$ , then

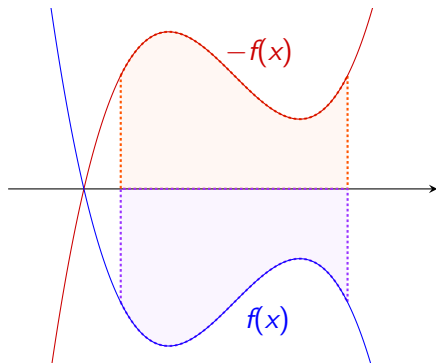
$$\int_a^b f(x) dx$$

is the area between the curve  $y = f(x)$  and the  $x$ -axis over the  $[a, b]$ .

- ▶ Consider now the case when  $f(x) \leq 0$  over  $[a, b]$ . Then  $-f(x) \geq 0$  and so, the area between the curve  $y = f(x)$  and the  $x$ -axis over the  $[a, b]$  is

$$\int_a^b -f(x) dx = - \int_a^b f(x) dx.$$

- ▶ This is necessary since area is nonnegative.



## In general

Find the area bounded by the graph of the function

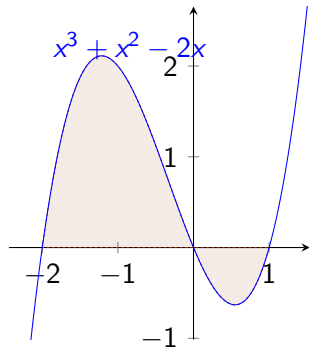
$$f(x) = x^3 + x^2 - 2x$$

and the  $x$ -axis, from  $-2$  to  $1$ .

The function is nonnegative on  $[-2, 0]$  and nonpositive on  $[0, 1]$ . Then means that  $\int_{-2}^0 f(x) dx \geq 0$  and  $\int_0^1 f(x) dx \leq 0$ .

So, the area is

$$\begin{aligned}\text{Area} &= \int_{-2}^0 f(x) dx - \int_0^1 f(x) dx = \int_{-2}^0 f(x) dx + \int_1^0 f(x) dx \\&= \left[ \frac{x^4}{4} + \frac{x^3}{3} - x^2 \right]_{-2}^0 + \left[ \frac{x^4}{4} + \frac{x^3}{3} - x^2 \right]_1^0 \\&= - \left[ \frac{(-2)^4}{4} + \frac{(-2)^3}{3} - (-2)^2 \right] - \left[ \frac{1^4}{4} + \frac{1^3}{3} - 1^2 \right] = \frac{37}{12}\end{aligned}$$



# Area Under Graph: Algorithm

Suppose  $f$  is continuous on  $[a, b]$ .

1. Find the **zeros** of  $f$ ,

$$\{x_1, x_2, \dots, x_n\}, \quad f(x_i) = 0 \quad \forall i = 1, \dots, n.$$

2. Determine the **sign** of  $f$  on each interval,

$$f(x) \geq 0 \quad \text{or} \quad f(x) \leq 0 \quad \text{on} \quad [x_i, x_{i+1}].$$

3. On the interval  $[x_i, x_{i+1}]$ , for  $i = 1, \dots, n - 1$ , define

$$f_i(x) = \begin{cases} f(x), & \text{if } f(x) \geq 0, \\ -f(x), & \text{if } f(x) \leq 0 \end{cases}$$

4. The **area** is the **sum**

$$\text{Area} = \int_a^b |f(x)| dx = \sum_{k=1}^{n-1} \int_{x_i}^{x_{i+1}} f_i(x) dx.$$

## Remark

Note that in the algorithm, it is important that  $f(x)$  is continuous on  $[a, b]$ .

1. A continuous function can only change sign by crossing the  $x$ -axis. This means every sign change corresponds to a zero of  $f$  so by locating the zeros we have indeed captured all the intervals where the function stays entirely above or below the axis.
2. On each subinterval between consecutive zeros, the function cannot “jump” from positive to negative or vice versa. So the sign of  $f$  is well-defined on each piece, and it makes sense to replace  $f$  by  $-f$  accordingly.
3. Without continuity,  $f$  can jump from negative to positive or vice versa without crossing the  $x$ -axis.

## Example

Find the area between the graph of

$$f(x) = \sin(2x)$$

and the  $x$ -axis, from  $-\pi$  to  $\pi$ .

$$\sin(2x) = 0 \quad \Leftrightarrow \quad 2x = -2\pi, -\pi, 0, \pi, 2\pi \quad \Leftrightarrow \quad x = -\pi, -\frac{\pi}{2}, 0, \frac{\pi}{2}, \pi.$$

$\sin(2x) \geq 0$  on  $[-\pi, -\frac{\pi}{2}]$  and  $[0, \frac{\pi}{2}]$ , and  $\sin(2x) \leq 0$  on  $[-\frac{\pi}{2}, 0]$  and  $[\frac{\pi}{2}, \pi]$ . Hence,

$$\begin{aligned} \text{Area} &= \int_{-\pi}^{\frac{\pi}{2}} f(x) \, dx - \int_{-\frac{\pi}{2}}^0 f(x) \, dx + \int_0^{\frac{\pi}{2}} f(x) \, dx - \int_{\frac{\pi}{2}}^{\pi} f(x) \, dx \\ &= \left[ -\frac{\cos(2x)}{2} \right]_{-\pi}^{\frac{\pi}{2}} - \left[ -\frac{\cos(2x)}{2} \right]_{-\frac{\pi}{2}}^0 + \left[ -\frac{\cos(2x)}{2} \right]_0^{\frac{\pi}{2}} - \left[ -\frac{\cos(2x)}{2} \right]_{\frac{\pi}{2}}^{\pi} \\ &= -\frac{1}{2} [-1 - 1 - 1 - 1 - 1 - 1 - 1 - 1] = 4 \end{aligned}$$

## Example

Alternatively, observe that  $|\sin(2x)|$  is an even function, so

$$\begin{aligned}\int_{-\pi}^{\pi} |\sin(2x)| dx &= 2 \int_0^{\pi} |\sin(2x)| dx \\ &= 2 \left( \int_0^{\frac{\pi}{2}} \sin(2x) dx - \int_{\frac{\pi}{2}}^{\pi} \sin(2x) dx \right) \\ &= 4\end{aligned}$$

One may even observe that since the period of  $|\sin(2x)|$  is  $\pi/2$ ,

$$\text{Area} = 4 \int_0^{\frac{\pi}{2}} \sin(2x) dx = 4.$$

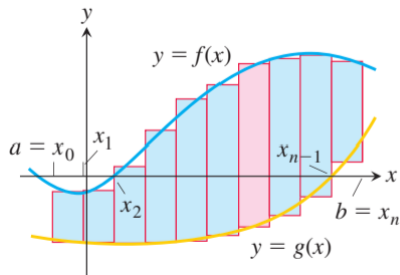
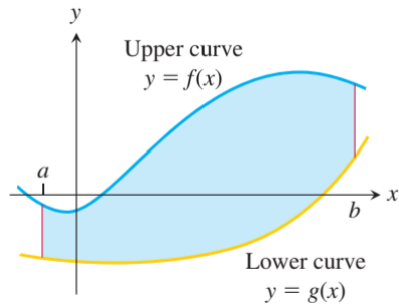
# Area Between Curves

- ▶ Suppose  $f(x)$  and  $g(x)$  are continuous on  $[a, b]$ .
- ▶ We want to find the area between the graphs of  $y = f(x)$  (upper curve) and  $y = g(x)$  (lower curve), from  $x = a$  to  $x = b$ .
- ▶ We approximate the area using vertical rectangles of width  $\Delta x_k$ :

$$\Delta A_k = \text{height} \times \text{width} = [f(c_k) - g(c_k)] \Delta x_k.$$

- ▶ Adding the areas of all  $n$  rectangles:

$$A \approx \sum_{k=1}^n [f(c_k) - g(c_k)] \Delta x_k.$$





# Area Between Curves

As the number of rectangles increases, the approximation becomes exact in the limit:

$$A = \lim_{n \rightarrow \infty} \sum_{k=1}^n [f(c_k) - g(c_k)] \Delta x_k = \int_a^b [f(x) - g(x)] dx.$$

## Definition

If  $f(x) \geq g(x)$  on  $[a, b]$ , then the area between the curves  $y = f(x)$  and  $y = g(x)$  is

$$A = \int_a^b [f(x) - g(x)] dx.$$

## Area Between Graph: Algorithm

Suppose  $f$  and  $g$  are continuous on  $[a, b]$ .

1. Find the **points of intersection** between  $f(x)$  and  $g(x)$  in  $[a, b]$ .

$$\{x_0 = a, x_1, \dots, x_n = b\}, \quad f(x_i) = g(x_i), \quad \forall i = 0, \dots, n.$$

2. Determine the **sign** of  $f(x) - g(x)$  on each interval  $[x_{i-1}, x_i]$ .
3. For  $i = 1, \dots, n$ , let

$$A_i = \begin{cases} \int_{x_{i-1}}^{x_i} (f(x) - g(x)) \, dx, & \text{if } f(x) \geq g(x) \\ \int_{x_{i-1}}^{x_i} (g(x) - f(x)) \, dx, & \text{if } g(x) \geq f(x) \end{cases}.$$

4. The **area** between the graph of  $f(x)$  and  $g(x)$  is

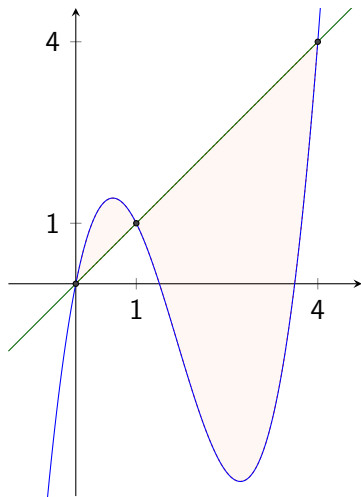
$$A = \sum_{i=1}^n A_i.$$

Alternatively, the **area** is

$$A = \int_a^b |f(x) - g(x)| \, dx.$$

## Example

Find the area between the graph of  $f(x) = x^3 - 5x^2 + 5x$  and  $g(x) = x$ , over  $[0, 4]$ .



$$f(x) = g(x) \Leftrightarrow f(x) - g(x) = x^3 - 5x^2 + 4x = 0 \Leftrightarrow x = 0, 1, 4.$$

$f(x) \geq g(x)$  over  $[0, 1]$  and  $g(x) \geq f(x)$  over  $[1, 4]$ . Hence, the area between the graphs is

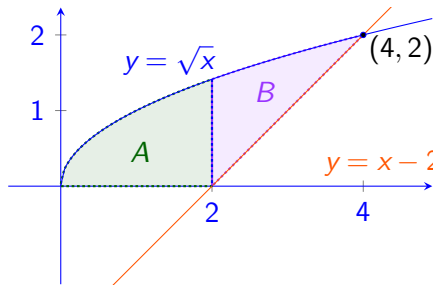
$$\begin{aligned} A &= \int_0^1 (f(x) - g(x)) \, dx + \int_1^4 (g(x) - f(x)) \, dx \\ &= \int_0^1 (x^3 - 5x^2 + 4x) \, dx - \int_1^4 (x^3 - 5x^2 + 4x) \, dx \\ &= \left[ \frac{x^4}{4} - \frac{5x^3}{3} + 2x^2 \right]_0^1 - \left[ \frac{x^4}{4} - \frac{5x^3}{3} + 2x^2 \right]_1^4 \\ &= \frac{1}{4} - \frac{5}{3} + 2 - \frac{4^4}{4} + \frac{5(4)^3}{3} - 2(4)^2 + \frac{1^4}{4} - \frac{5}{3} + 2 \\ &= \frac{71}{6} \end{aligned}$$

## Example

Find the area of the region in the first quadrant that is bounded above by  $y = \sqrt{x}$  and below by the  $x$ -axis and the line  $y = x - 2$ .

Let  $f(x) = \sqrt{x}$  and  $g(x) = x - 2$ . The total area is

$$\begin{aligned} \int_0^2 f(x) dx + \int_2^4 f(x) - g(x) dx &= \int_0^4 f(x) dx - \int_2^4 g(x) dx \\ &= \left[ \frac{2}{3} x^{\frac{3}{2}} \right]_0^4 - \left[ \frac{x^2}{2} - 2x \right]_2^4 = \frac{16}{3} - 2 \\ &= \frac{10}{3}. \end{aligned}$$

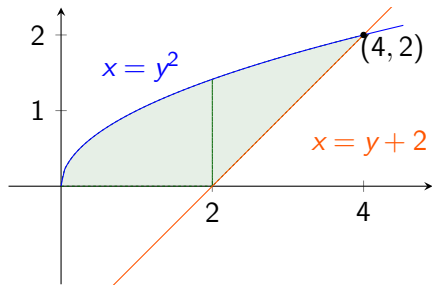


## Integration With Respect to y

Due to the nature of the area, or for shorter computation, one might have to consider the curve as graph of functions of  $y$ .

For example, for the previous example, we may instead let  $f(y) = y + 2$  and  $g(y) = y^2$ . Then the area is

$$\begin{aligned}\int_0^2 (f(y) - g(y)) dy &= \left[ \frac{y^2}{2} + 2y - \frac{y^3}{3} \right]_0^2 \\ &= 2 + 4 - \frac{8}{3} \\ &= \frac{10}{3}.\end{aligned}$$

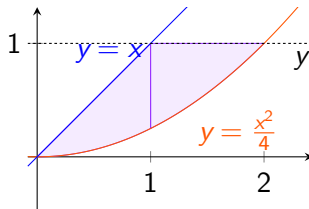


## Example

Find the area bounded between the line  $y = 1$ , and the graphs  $y = x$  and  $y = \frac{x^2}{4}$ .

Let  $f(y) = 2\sqrt{y}$  and  $g(y) = y$ . Then the area is

$$\begin{aligned}\int_0^1 (f(y) - g(y)) dy &= \left[ \frac{4}{3} y^{\frac{3}{2}} - \frac{y^2}{2} \right]_0^1 \\ &= \frac{4}{3} - \frac{1}{2} \\ &= \frac{5}{6}.\end{aligned}$$



# Integration by Substitution

# Introduction

Find the indefinite integral

$$\int x\sqrt{2x+1}dx$$

- (i) No obvious antiderivative.
- (ii) Try to use power rule, but by product rule,

$$\frac{d}{dx} \left( x^2(2x+1)^{\frac{3}{2}} \right) = 2x(2x+1)^{\frac{3}{2}} + \frac{3}{2}x^2\sqrt{2x+1}.$$

- (iii) How about

$$\frac{d}{dx} \left( \frac{2}{3}x(2x+1)^{\frac{3}{2}} \right) = \frac{2}{3}(2x+1)^{\frac{3}{2}} + x\sqrt{2x+1}$$

extra term:  $\frac{2}{3}(2x+1)^{\frac{3}{2}}.$



## Introduction

Consider another example. Find the indefinite integral of

$$\int (x^3 + x)^5 (3x^2 + 1) dx.$$

Consider the term  $(x^3 + x)^6$ . By chain rule,

$$\frac{d}{dx}(x^3 + x)^6 = 6(x^3 + x)^5(3x^2 + 1).$$

So,  $\frac{(x^3 + x)^6}{6}$  is an antiderivative!

Idea: Let  $u(x) = x^3 + x$ . Then  $\frac{d}{dx}u = (3x^2 + 1)$  and the integration becomes

$$\int u(x)^5 u'(x) dx \text{ or } \int u^5 \frac{du}{dx} dx.$$

# Integration by Substitution: Indefinite integral

## Proposition

If the *integral* is of the form

$$\int f(u(x))u'(x) dx$$

for some  $u(x)$ , then for any *antiderivative*  $F$  of  $f$ ,

$$\int f(u(x))u'(x) dx = F(u(x)) + C.$$

Proof.

By the fundamental theorem of calculus, we may directly check that by chain rule,

$$\frac{d}{dx}(F(u(x)) + C) = F'(u(x))u'(x) = f(u(x))u'(x)$$

since  $F'(y) = f(y)$ .



## Remarks

1. Write  $u'(x) = \frac{du}{dx}$  we may treat it as a fraction and thus  $\frac{du}{dx}dx = du$ . Then the substitution formula becomes

$$\int f(u(x)) \frac{du}{dx} dx = \int f(u) du = F(u) + C.$$

2. Implicitly, the proposition assumes that the function  $u$  is **differentiable** (on some interval).
3. Also, the function  $f$  must have easily identified **antiderivative**, and is **continuous** on the range of  $u$ .
4. The technique will be as follows.
  - (a) Identify  $u(x)$  by checking that the integrand contains the term  $u'(x)$ .
  - (b) Identify the function  $f$  such that the integrand is

$$f(u(x))u'(x).$$

- (c) Find an antiderivative  $F$  of  $f$ .

## Example

Find the indefinite integral

$$\int x\sqrt{x^2 + 1} dx.$$

Let  $u = x^2 + 1$ ,  $f(u) = \sqrt{u}$ . Then

$$F(u) = \frac{2}{3}u^{\frac{3}{2}} \quad \text{and} \quad u'(x) = 2x.$$

The integrand does not contain the term  $u'(x) = 2x$ . We can easily fix this.

$$\begin{aligned} \int x\sqrt{x^2 + 1} dx &= \frac{1}{2} \int 2x\sqrt{x^2 + 1} dx = \frac{1}{2} \int \sqrt{x^2 + 1} (2x) dx \\ &= \frac{1}{2} \int \sqrt{u} du = \frac{1}{2} \left( \frac{2}{3} u^{\frac{3}{2}} \right) + C \\ &= \frac{1}{3} (x^2 + 1)^{\frac{3}{2}} + C \end{aligned}$$

## Example

Find

$$\int \sin(ax + b) dx$$

for some fixed  $a, b \in \mathbb{R}$ ,  $a \neq 0$ .

Let  $u(x) = ax + b$ ,  $f(u) = \sin(u)$ . Then

$$F(u) = -\cos(u) \quad \text{and} \quad du = adx.$$

Hence,

$$\begin{aligned} \int \sin(ax + b) dx &= \frac{1}{a} \int \sin(ax + b) adx = \frac{1}{a} \int \sin(u) du \\ &= -\frac{1}{a} \cos(u) + C \\ &= -\frac{\cos(ax + b)}{a} + C. \end{aligned}$$

## Example

Find

$$\int \frac{x}{\sqrt{1-4x^2}} dx.$$

Let  $u(x) = 1 - 4x^2$ .  $u'(x) = -8x$ . Then

$$\begin{aligned} \int \frac{x}{\sqrt{1-4x^2}} dx &= -\frac{1}{8} \int \frac{-8x dx}{\sqrt{1-4x^2}} = -\frac{1}{8} \int \frac{du}{\sqrt{u}} \\ &= -\frac{1}{8} (2\sqrt{u}) + C = -\frac{\sqrt{1-4x^2}}{4} + C. \end{aligned}$$

## Example

Find

$$\int x^5 \sqrt{1+x^2} dx.$$

Let  $u(x) = 1 + x^2$ , then  $u'(x) = 2x$ . Now  $x^2 = u - 1 \implies x^4 = (u - 1)^2$ , so

$$x^5 = \frac{1}{2}(2x)x^4 = \frac{1}{2}u'(x)(u(x) - 1)^2.$$

So,

$$\begin{aligned} \int x^5 \sqrt{1+x^2} dx &= \frac{1}{2} \int \sqrt{u(x)}(u(x) - 1)^2 u'(x) dx = \frac{1}{2} \int \sqrt{u}(u - 1)^2 du = \frac{1}{2} \int \sqrt{u}(u^2 - 2u + 1) du \\ &= \frac{1}{2} \int u^{\frac{5}{2}} - 2u^{\frac{3}{2}} + u^{\frac{1}{2}} du = \frac{1}{7} u^{\frac{7}{2}} - \frac{2}{5} u^{\frac{5}{2}} + \frac{1}{3} u^{\frac{3}{2}} + C \\ &= \frac{(1+x^2)^{\frac{7}{2}}}{7} - \frac{2(1+x^2)^{\frac{5}{2}}}{5} + \frac{(1+x^2)^{\frac{3}{2}}}{3} + C. \end{aligned}$$

## Discussion

Back to the first example. Find

$$\int x\sqrt{2x+1}dx$$

Could we use the substitution  $u(x) = \sqrt{2x+1}$  to solve?

$$u(x) = \sqrt{2x+1}, \quad u^2 = 2x+1, \quad (2u(x))u'(x) = 2 \quad \text{and} \quad x = \frac{1}{2}(u(x)^2 - 1).$$

So,

$$\begin{aligned} \int x\sqrt{2x+1}dx &= \int \frac{1}{2}(u(x)^2 - 1)u(x)u'(x) dx = \frac{1}{2} \int (u^2 - 1)u^2 du = \frac{1}{2} \int u^4 - u^2 du \\ &= \frac{1}{2} \left[ \frac{u^5}{5} - \frac{u^3}{3} \right] + C \\ &= \frac{(2x+1)^{\frac{5}{2}}}{10} - \frac{(2x+1)^{\frac{3}{2}}}{6} + C \end{aligned}$$



# Integration by Substitution: Definite Integral

Evaluate

$$\int_0^1 x\sqrt{x^2+1}dx.$$

As before,  $u(x) = x^2 + 1$ ,  $u'(x) = 2x$ . Then

$$\int_0^1 x\sqrt{x^2+1}dx = \left[ \frac{1}{3} (x^2+1)^{\frac{3}{2}} \right]_0^1 = \frac{2^{\frac{3}{2}} - 1}{3}.$$

In details,

$$\begin{aligned} \int_0^1 x\sqrt{x^2+1}dx &= \frac{1}{2} \int_0^1 \sqrt{x^2+1}(2x) dx = \frac{1}{2} \int_{x=0}^{x=1} \sqrt{u} du \\ &= \left[ \frac{1}{3} u^{\frac{3}{2}} \right]_{x=0}^{x=1} \quad u(1) = 2, \quad u(0) = 1 \\ &= \frac{1}{3} \left( 2^{\frac{3}{2}} - 1 \right) \end{aligned}$$

# Integration by Substitution: Definite Integral

Evaluate

$$\int_0^1 x\sqrt{x^2 + 1}dx.$$

Could have substitute  $u(1) = 2$ ,  $u(0) = 1$  when changing variable.

$$\int_0^1 x\sqrt{x^2 + 1}dx = \frac{1}{2} \int_{u=1}^{u=2} \sqrt{u}du = \left[ \frac{1}{3} u^{\frac{3}{2}} \right]_{u=1}^{u=2} = \frac{1}{3} \left( 2^{\frac{3}{2}} - 1 \right).$$

# Integration by Substitution: Definite Integral

## Proposition

If  $u'$  is *continuous* on the interval  $[a, b]$  and  $f$  is *continuous* on the *range* of  $u$ , then

$$\int_a^b f(u(x))u'(x) \, dx = \int_{u(a)}^{u(b)} f(u) \, du.$$

Proof.

Let  $F$  be any antiderivative of  $f$ .

$$\begin{aligned} \int_a^b f(u(x))u'(x) \, dx &= [F(u)]_{x=a}^{x=b} = F(u(b)) - F(u(a)) \\ &= [F(u)]_{u(a)}^{u(b)} = \int_{u(a)}^{u(b)} f(u) \, du. \end{aligned}$$



## Example

$$\int_{\ln(2)}^{\ln(3)} \frac{1}{e^x + e^{-x}} dx.$$

First observe that

$$\frac{1}{e^x + e^{-x}} = \frac{e^x}{e^{2x} + 1}.$$

Let  $u(x) = e^x$ . Then  $u'(x) = e^x$ ,  $u(\ln(2)) = 2$ , and  $u(\ln(3)) = 3$ .

$$\begin{aligned} \int_{\ln(2)}^{\ln(3)} \frac{1}{e^x + e^{-x}} dx &= \int_{\ln(2)}^{\ln(3)} \frac{e^x}{e^{2x} + 1} dx = \int_2^3 \frac{1}{u^2 + 1} du \\ &= [\tan^{-1}(u)]_2^3 \\ &= \tan^{-1}(3) - \tan^{-1}(2). \end{aligned}$$

## Example

$$\int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \tan(x) \, dx$$

Since  $\tan(x) = \frac{\sin(x)}{\cos(x)}$ , let  $u(x) = \cos(x)$ . Then  $u'(x) = -\sin(x)$ ,  $u(-\frac{\pi}{4}) = -\frac{\sqrt{2}}{2}$ , and  $u(\frac{\pi}{4}) = \frac{\sqrt{2}}{2}$ . Hence,

$$\begin{aligned} \int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \tan(x) \, dx &= \int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} -\frac{u'(x)}{u(x)} \, dx = \int_{-\frac{\sqrt{2}}{2}}^{\frac{\sqrt{2}}{2}} \frac{du}{u} \\ &= [\ln(|u|)]_{-\frac{\sqrt{2}}{2}}^{\frac{\sqrt{2}}{2}} \\ &= 0. \end{aligned}$$

## Second Substitution Rule

Let  $x = x(t)$  be a differentiable function of  $t$ . Then:

$$\int f(x) dx = \int f(x(t)) \cdot \frac{dx}{dt} dt.$$

This allows us to change variables when it is easier to express the integral in terms of a new variable  $t$ .

Compare:

- ▶ First substitution:  $u = u(x)$ , reduce  $\int f(u(x))u'(x) dx$  to  $\int f(u) du$ .
- ▶ Second substitution:  $x = x(t)$ , rewrite  $\int f(x) dx$  in terms of  $t$  using  $x(t)$ .

## Example

Evaluate

$$\int \frac{1}{\sqrt{x}(x+1)} dx.$$

Let  $t = \sqrt{x}$ , so  $x = t^2$  and  $\frac{dx}{dt} = 2t$ .

$$\int \frac{1}{\sqrt{x}(x+1)} dx = \int \frac{1}{t(t^2+1)} \cdot 2t dt = 2 \int \frac{1}{t^2+1} dt = 2 \tan^{-1}(t) + C = 2 \tan^{-1}(\sqrt{x}) + C.$$

# Trigonometric Substitution

Evaluate

$$\int \frac{dx}{\sqrt{1+x^2}}.$$

Let  $x(\theta) = \tan \theta$ , and define  $f(x) = \frac{1}{\sqrt{1+x^2}}$ . Then:

$$f(x(\theta)) = \frac{1}{\sqrt{1+\tan^2 \theta}} = \frac{1}{\sec \theta} = \cos \theta,$$

and

$$x'(\theta) = \frac{d}{d\theta}(\tan \theta) = \sec^2 \theta.$$

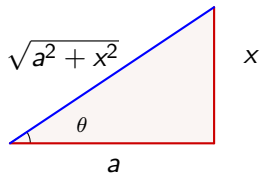
So,

$$\int \frac{dx}{\sqrt{1+x^2}} = \int \cos \theta \cdot \sec^2 \theta d\theta = \int \sec \theta d\theta = \ln |\sec \theta + \tan \theta| + C.$$

Express it back in terms of  $x$ :  $\ln |x + \sqrt{1+x^2}| + C.$

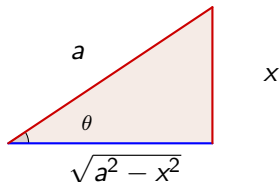


# Trigonometric Substitution Triangle



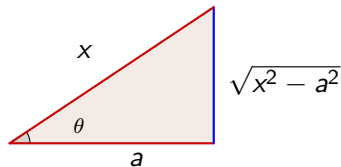
$$x = a \tan \theta$$

$$\begin{aligned}\sqrt{a^2 + x^2} &= \sqrt{a^2 + a^2 \tan^2 \theta} \\ &= a\sqrt{1 + \tan^2 \theta} \\ &= a \sec \theta\end{aligned}$$



$$x = a \sin \theta$$

$$\begin{aligned}\sqrt{a^2 - x^2} &= \sqrt{a^2 - a^2 \sin^2 \theta} \\ &= a\sqrt{1 - \sin^2 \theta} \\ &= a \cos \theta\end{aligned}$$



$$x = a \sec \theta$$

$$\begin{aligned}\sqrt{x^2 - a^2} &= \sqrt{a^2 \sec^2 \theta - a^2} \\ &= a\sqrt{\sec^2 \theta - 1} \\ &= a \tan \theta\end{aligned}$$

## Example

Evaluate

$$\int \frac{dx}{\sqrt{a^2 + x^2}}.$$

Let  $x(\theta) = a \tan \theta$ . Then  $x'(\theta) = a \sec^2 \theta$ , and

$$\frac{1}{\sqrt{a^2 + x(\theta)^2}} = \frac{1}{\sqrt{a^2 + a^2 \tan^2 \theta}} = \frac{1}{a \sec \theta}.$$

Hence,

$$\int \frac{dx}{\sqrt{a^2 + x^2}} = \int \frac{1}{a \sec \theta} \cdot a \sec^2 \theta d\theta = \int \sec \theta d\theta = \ln |\sec \theta + \tan \theta| + C.$$

## Example

Evaluate

$$\int \frac{x^2 dx}{\sqrt{9-x^2}}$$

Let  $x(\theta) = 3 \sin(\theta)$ . Then  $x'(\theta) = 3 \cos(\theta)$  and

$$\frac{x(\theta)^2}{\sqrt{9-x(\theta)^2}} = \frac{9 \sin^2(\theta)}{\sqrt{9-9 \sin^2(\theta)}} = \frac{3 \sin^2(\theta)}{\cos(\theta)}.$$

Hence,

$$\begin{aligned} \int \frac{x^2 dx}{\sqrt{9-x^2}} &= \int \frac{3 \sin^2(\theta)}{\cos(\theta)} 3 \cos(\theta) d\theta = 9 \int \sin^2(\theta) d\theta = 9 \int \frac{1 - \cos(2\theta)}{2} d\theta \\ &= \frac{9}{2} \theta - \frac{9}{4} \sin(2\theta) + C = \frac{9}{2} \theta - \frac{9}{2} (\sin(\theta) \cos(\theta)) + C \\ &= \frac{9}{2} \sin^{-1} \left( \frac{x}{3} \right) - \frac{x \sqrt{9-x^2}}{2} + C \end{aligned}$$

## Example

Evaluate

$$\int \frac{dx}{(1+x^2)^3}.$$

Let  $x(\theta) = \tan(\theta)$ . Then  $x'(\theta) = \sec^2(\theta)$  and

$$\frac{1}{(1+x(\theta)^2)^3} = \frac{1}{(1+\tan^2(\theta))^3} = \frac{1}{\sec^6(\theta)}.$$

Also,  $\sin(\theta) = \frac{x}{\sqrt{1+x^2}}$  and  $\cos(\theta) = \frac{1}{\sqrt{1+x^2}}$ . Hence,

$$\begin{aligned} \int \frac{dx}{(1+x^2)^3} &= \int \frac{\sec^2(\theta)}{\sec^6(\theta)} d\theta = \int \cos^4(\theta) d\theta = \frac{1}{4} \sin(\theta) \cos^3(\theta) + \frac{3}{8} \sin(\theta) \cos(\theta) + \frac{3}{8} \theta + C \\ &= \frac{1}{4} \frac{x}{\sqrt{1+x^2}} \frac{1}{\sqrt{1+x^2}^3} + \frac{3}{8} \frac{x}{\sqrt{1+x^2}} \frac{1}{\sqrt{1+x^2}} + \frac{3}{8} \tan^{-1}(x) + C \\ &= \frac{1}{4} \frac{x}{(1+x^2)^2} + \frac{3}{8} \frac{x}{1+x^2} + \frac{3}{8} \tan^{-1}(x) + C. \end{aligned}$$

## Example

Evaluate

$$\int x\sqrt{5+4x-x^2}dx.$$

By completing the square,  $5+4x-x^2 = 3^2 - (x-2)^2$ .

Let  $x(\theta) - 2 = 3\sin(\theta)$ , that is,  $x(\theta) = 3\sin(\theta) + 2$ . Then  $x'(\theta) = 3\cos(\theta)$  and

$$x(\theta)\sqrt{5+4x(\theta)-x(\theta)^2} = (3\sin(\theta) + 2)\sqrt{3^2 - 3^2\sin^2(\theta)} = (3\sin(\theta) + 2)(3\cos(\theta)).$$

Also,  $\cos(\theta) = \frac{\sqrt{3^2 - (x-2)^2}}{3} = \frac{\sqrt{5+4x-x^2}}{3}$ . Hence,

$$\begin{aligned}\int x\sqrt{5+4x-x^2}dx &= \int (3\sin(\theta) + 2)(3\cos(\theta))^2 d\theta = 27 \int \sin(\theta) \cos^2(\theta) d\theta + 18 \int \cos^2(\theta) d\theta \\ &= -27 \int u^2 du + 9\sin(\theta)\cos(\theta) + 9\theta + C = -9\frac{u^3}{3} + 9\sin(\theta)\cos(\theta) + 9\theta + C \\ &= -\frac{(5+4x-x^2)^{\frac{3}{2}}}{3} + (x-2)\sqrt{5+4x-x^2} + 9\sin^{-1}\left(\frac{x-2}{3}\right) + C\end{aligned}$$

# The Natural Logarithm and Exponential Function

# Introduction

- ▶ We have been using and working with the natural logarithm  $\ln(x)$  and natural exponential function  $e^x$ , together with their algebraic properties.
- ▶ Although  $\ln(x) = \log_e(x)$  was introduced as the inverse of  $e^x$ , we have not yet given a precise definition of the constant  $e \approx 2.718$ .
- ▶ With the Fundamental Theorem of Calculus available, we now adopt a more foundational approach: we define  $\ln(x)$  directly as an integral.
- ▶ From this integral definition, we then define the number  $e$  naturally as the unique number satisfying  $\ln(e) = 1$ .
- ▶ Finally, we establish rigorously that the exponential function  $e^x$  is indeed the inverse of the logarithmic function  $\ln(x)$ .

# Definition of $\ln(x)$

## Definition

The natural logarithm is the function given by

$$\ln(x) := \int_1^x \frac{1}{t} dt, \quad x > 0.$$

- ▶ From the Fundamental Theorem of Calculus,  $\ln(x)$  is continuous since  $\frac{1}{t}$  is.
- ▶ Geometrically, if  $x > 1$ ,  $\ln(x)$  is the area under the curve  $\frac{1}{t}$  from  $t = 1$  to  $t = x$ . For  $0 < x < 1$ ,  $\ln(x)$  gives the negative area under the curve from  $t = x$  to  $t = 1$ .
- ▶ By the zero width interval property,

$$\ln(1) = \int_1^1 \frac{1}{t} dt = 0.$$



## Derivative of $\ln(x)$

By the Fundamental Theorem of Calculus,

$$\frac{d}{dx} \ln(x) = \frac{d}{dx} \int_1^x \frac{1}{t} dt = \frac{1}{x}$$

for all  $x \geq 0$ .

Consider the function  $u(x) = |x|$ . For all  $x \neq 0$ ,  $u(x)$  is differentiable and  $\frac{d}{dx} u(x) = \frac{x}{|x|}$ . Hence, by the chain rule,

$$\frac{d}{dx} \ln |x| = \frac{d}{du} \ln(u) \frac{du}{dx} = \frac{1}{u} \cdot \frac{x}{|x|} = \frac{1}{|x|} \cdot \frac{x}{|x|} = \frac{x}{x^2} = \frac{1}{x}.$$

Therefore,

$$\int \frac{1}{x} dx = \ln |x| + C.$$

## Power Rule for $\ln(x)$

We have defined  $\ln(x) = \int_1^x \frac{1}{t} dt$ . From this definition alone, the power rule,  $\ln(x^k) = k \cdot \ln(x)$  for any real number  $k$ , does not follow automatically. We now establish this property using substitution.

Proof.

Consider  $\ln(x^k) = \int_1^{x^k} \frac{1}{t} dt$ . Then  $dt = k \cdot u^{k-1} du$ . Use (second) substitution  $u^k = t$ ,

- ▶ when  $t = 1$ ,  $u = 1$ ;
- ▶ when  $t = x^k$ ,  $u = x$ .

Hence,

$$\begin{aligned}\ln(x^k) &= \int_1^{x^k} \frac{1}{t} dt = \int_1^x \frac{1}{u^k} \cdot k \cdot u^{k-1} du \\ &= k \int_1^x \frac{u^{k-1}}{u^k} du = k \int_1^x \frac{1}{u} du \\ &= k \ln(x).\end{aligned}$$

# Definition of $e$

## Definition

The natural number  $e$  is the number in the domain of the natural logarithm that satisfies

$$\ln(e) = \int_1^e \frac{1}{t} dt = 1.$$

That is,  $e$  is defined to be the number such that the area under the graph of  $\frac{1}{t}$  from  $t = 1$  to  $e$  is exactly 1 unit<sup>2</sup>.

## Proof of $\log_e = \ln$

Now since  $\frac{1}{t}$  is continuous and increasing for  $t > 0$ , then so is  $\ln(x) = \int_1^x \frac{1}{t} dt$ .

Thus,  $\ln(x)$  is **injective** (**one-to-one**) with domain  $(0, \infty)$  and range  $(-\infty, \infty)$ , and there has an inverse, which we denote by  $\ln^{-1}(x)$ .

Let  $e$  be the natural number (defined by  $\ln(e) = 1$ ). Consider the exponential function  $e^x$ . It is a continuous and strictly increasing exponential function, so it has an inverse  $\log_e(x)$ . We will now show that  $\log_e(x) = \ln(x)$ .

Using the power rule for  $\ln$ , for any real  $x$ ,

$$\ln(e^x) = x \cdot \ln(e) = x$$

since  $\ln(e) = 1$  by definition. Apply the inverse of  $\ln(x)$  on both sides gives

$$e^x = \ln^{-1}(x),$$

which shows that  $e^x$  is the inverse of  $\ln(x)$ . Hence, by uniqueness of inverse of  $e^x$ ,

$$\ln = \log_e$$

## Integration by Parts

# Introduction

Recall the **product rule** for differentiation

$$\frac{d}{dx}(u(x)v(x)) = u'(x)v(x) + u(x)v'(x).$$

By the **fundamental theorem of calculus**,

$$\int (u'(x)v(x) + u(x)v'(x)) dx = u(x)v(x) + C,$$

which can be rearrange to be written as

$$\int u(x)v'(x) dx = u(x)v(x) - \int u'(x)v(x) dx + C.$$

This is called the **integration by parts** formula. In **differential form**, it can be written as

$$\int u dv = uv - \int v du.$$

## Examples

1. Find  $\int x \sin(x) dx$ .

Let  $u(x) = x$  and  $v'(x) = \sin(x)$ . Then  $u'(x) = 1$  and  $v(x) = -\cos(x)$ . So,

$$\int x \sin(x) dx = -x \cos(x) + \int (1) \cos(x) dx = \sin(x) - x \cos(x) + C.$$

2. Find  $\int \ln(x) dx$ .

There is no known antiderivative of  $\ln(x)$ . But we know its derivative. Try let  $u(x) = \ln(x)$  and  $v'(x) = 1$ . Then  $u'(x) = \frac{1}{x}$  and  $v(x) = x$ . Hence,

$$\int \ln(x) dx = x \ln(x) - \int \frac{1}{x} \cdot x dx = x \ln(x) + x + C.$$

## Examples

3. Find  $\int x^2 e^x dx$ .

Try  $u(x) = x^2$  and  $v(x) = e^x$ . Then  $u'(x) = 2x$  and  $v(x) = e^x$ . So,

$$\int x^2 e^x dx = x^2 e^x - 2 \int x e^x dx$$

No obvious antiderivative for  $x e^x$ . Perform integration by parts again. This time,  $u(x) = x$  and  $v(x) = e^x$ . Then  $u'(x) = 1$  and  $v(x) = e^x$ . So,

$$\begin{aligned} \int x^2 e^x dx &= x^2 e^x - 2 \int x e^x dx = x^2 e^x - 2 \left( x e^x - \int e^x dx \right) \\ &= e^x (x^2 - 2x + 1) \\ &= e^x (x - 1)^2. \end{aligned}$$



## Examples

4. Find  $\int \sin^{-1}(x) dx$ .

No obvious antiderivative for  $\sin^{-1}(x)$ . But we do know its derivative.  
Let  $u(x) = \sin^{-1}(x)$  and  $v(x) = 1$ . Then  $u'(x) = \frac{1}{\sqrt{1-x^2}}$  and  $v(x) = x$ .

$$\int \sin^{-1}(x) dx = x \sin^{-1}(x) - \int \frac{x}{\sqrt{1-x^2}} dx.$$

May solve the second integral using substitution. Let (abuse of notations)  $u(x) = 1 - x^2$  (poor  $u$ ).  
Then  $u'(x) = -2x$ . So,

$$\begin{aligned} \int \sin^{-1}(x) dx &= x \sin^{-1}(x) - \int \frac{x}{\sqrt{1-x^2}} dx = x \sin^{-1}(x) + \frac{1}{2} \int \frac{-2x}{\sqrt{1-x^2}} dx \\ &= x \sin^{-1}(x) + \frac{1}{2} \int u^{-\frac{1}{2}} du = x \sin^{-1}(x) + \frac{1}{2} (2u^{\frac{1}{2}}) + C \\ &= x \sin^{-1}(x) + \sqrt{1-x^2} + C. \end{aligned}$$

## Examples

5. Find  $\int e^x \sin(x) dx$ .

Let  $u(x) = \sin(x)$  and  $v(x) = e^x$ . Then  $u'(x) = \cos(x)$  and  $v(x) = e^x$ . So,

$$\int e^x \sin(x) dx = e^x \sin(x) - \int e^x \cos(x) dx.$$

Try integration by parts again.  $u(x) = \cos(x)$ , same  $v$  as above. Then  $u'(x) = -\sin(x)$  and so

$$\int e^x \sin(x) dx = e^x \sin(x) - \int e^x \cos(x) dx = e^x \sin(x) - e^x \cos(x) - \int e^x \sin(x) dx$$

Seems like we are going in circle. But notice the term  $\int e^x \sin(x) dx$  on both sides. Rearranging, we have

$$\int e^x \sin(x) dx = \frac{1}{2} e^x (\sin(x) - \cos(x)) + C.$$

# Integration of Rational Functions by Partial Fractions

# Introduction

Consider the

$$\int \frac{5x - 3}{x^2 - 2x - 3} dx.$$

By writing

$$\frac{5x - 3}{x^2 - 2x - 3} = \frac{2}{x + 1} + \frac{3}{x - 3},$$

we can compute the integral easily,

$$\int \frac{5x - 3}{x^2 - 2x - 3} dx = \int \left( \frac{2}{x + 1} + \frac{3}{x - 3} \right) dx = 2 \ln |x + 1| + 3 \ln |x - 3| + C$$

The method of rewriting **rational functions** as sums of simpler fractions is called the **method of partial fractions**.

# Partial Fractions

Consider a **rational fraction**

$$f(x) = \frac{p(x)}{q(x)},$$

where  $p(x)$  and  $q(x)$  are **polynomials**.

1. First write  $f(x)$  as a **mixed fraction**,

$$f(x) = \frac{p_1(x)}{q_1(x)} + r(x),$$

where  $\deg(p_1) < \deg(q_1)$ , if necessary. Henceforth in the discussion, we will assume that  $f(x) = \frac{p(x)}{q(x)}$  is a **proper fraction**.

2. For the discussion on **partial fractions**, we will consider the cases where the **polynomial**  $q(x)$  has **linear factors**, or **irreducible quadratic factors**,

$$q(x) = (x - d_1)^{r_1} \cdots (x - d_k)^{r_k} (a_1 x^2 + b_1 x + c_1)^{s_1} \cdots (a_l x^2 + b_l x + c_l)^{s_l},$$

where  $(x - d_i)$ ,  $i = 1, \dots, k$ , and  $(a_j x^2 + b_j x + c_j)$ ,  $j = 1, \dots, l$  are **distinct** factors.

## Partial Fractions

3. For each **factor**  $(x - d_i)^{r_i}$ , we may write it as

$$\frac{1}{(x - d_i)^{r_i}} = \frac{A_1}{(x - d_i)} + \frac{A_2}{(x - d_i)^2} + \cdots + \frac{A_{r_i}}{(x - d_i)^{r_i}}.$$

4. For each **factor**  $(a_j x^2 + b_j x + c_j)^{s_j}$ , we may write it as

$$\frac{1}{(a_j x^2 + b_j x + c_j)^{s_j}} = \frac{B_1 x + C_1}{(a_j x^2 + b_j x + c_j)} + \frac{B_2 x + C_2}{(a_j x^2 + b_j x + c_j)^2} + \cdots + \frac{B_{s_j} x + C_{s_j}}{(a_j x^2 + b_j x + c_j)^{s_j}}.$$

5. Set the **rational fraction** as a sum of the **partial fractions** above, and solve for the unknown  $A$ 's,  $B$ 's, and  $C$ 's.

**Illustration:** The irreducible polynomials behaves like prime numbers in the integers. Here is an example of the method of partial fractions illustrated in integers.

$$\frac{7}{45} = \frac{7}{(3)^2(5)} = \frac{1}{3} + \frac{2}{3^2} - \frac{2}{5}.$$

## Example

Evaluate

$$\int \frac{x^2 + 2x - 1}{2x^3 + 3x^2 - 2x} dx.$$

►  $\frac{x^2 + 2x - 1}{2x^3 + 3x^2 - 2x}$  is a proper fraction.

► Decompose the denominator:  $2x^3 + 3x^2 - 2x = x(2x^2 + 3x - 2) = x(x + 2)(2x - 1)$ .

$$\frac{x^2 + 2x - 1}{2x^3 + 3x^2 - 2x} = \frac{A}{x} + \frac{B}{x+2} + \frac{C}{2x-1} = \frac{A(x+2)(2x-1) + Bx(2x-1) + Cx(x+2)}{x(x+2)(2x-1)}$$

$$\begin{aligned} &= \frac{(2A + 2B + C)x^2 + (3A - B + 2C)x - 2A}{x(x+2)(2x-1)} \\ &\Rightarrow A = \frac{1}{2}, \quad B = \frac{-1}{10}, \quad C = \frac{1}{5}. \end{aligned}$$

$$\begin{aligned} \int \frac{x^2 + 2x - 1}{2x^3 + 3x^2 - 2x} dx &= \int \left( \frac{1}{2x} - \frac{1}{10(x+2)} + \frac{1}{5(2x-1)} \right) dx \\ &= \frac{1}{2} \ln|x| - \frac{1}{10} \ln|x+2| + \frac{1}{10} \ln|2x-1| + C. \end{aligned}$$

## Example

Evaluate

$$\int \frac{2x^2 - x + 4}{x^3 + 4x} dx.$$

►  $\frac{2x^2 - x + 4}{x^3 + 4x}$  is a proper fraction.  $x^3 + 4x = x(x^2 + 4)$ .

►

$$\begin{aligned} \frac{2x^2 - x + 4}{x^3 + 4x} &= \frac{A}{x} + \frac{Bx + C}{x^2 + 4} = \frac{A(x^2 + 4) + (Bx + C)x}{x(x^2 + 4)} = \frac{(A + B)x^2 + Cx + 4A}{x(x^2 + 4)} \\ &\Rightarrow A = 1, \quad B = 1, \quad C = -1. \end{aligned}$$

►

$$\int \frac{2x^2 - x + 4}{x^3 + 4x} dx = \int \left( \frac{1}{x} + \frac{x}{x^2 + 4} - \frac{1}{x^2 + 4} \right) dx = \ln |x| + \frac{1}{2} \ln |x^2 + 4| - \tan^{-1} \left( \frac{x}{2} \right) + C.$$



## Example

Evaluate

$$\int \frac{x^4 - 2x^2 + 4x + 1}{x^3 - x^2 - x + 1} dx.$$

► By long division,  $\frac{x^4 - 2x^2 + 4x + 1}{x^3 - x^2 - x + 1} = (x + 1) + \frac{4x}{x^3 - x^2 - x + 1}$ .  $x^3 - x^2 - x + 1 = (x + 1)(x - 1)^2$ .

►

$$\frac{4x}{x^3 - x^2 - x + 1} = -\frac{1}{x + 1} + \frac{1}{x - 1} + \frac{2}{(x - 1)^2}$$

Exercise: Work out the derivations.

►

$$\begin{aligned} \int \frac{x^4 - 2x^2 + 4x + 1}{x^3 - x^2 - x + 1} dx &= \int \left( x + 1 - \frac{1}{x + 1} + \frac{1}{x - 1} + \frac{2}{(x - 1)^2} \right) dx \\ &= \frac{x^2}{2} + x - \ln |x + 1| + \ln |x - 1| - \frac{2}{x - 1} + C \end{aligned}$$

# Improper Integral

# Introduction

Consider the following **definite integral**

$$\int_a^b \frac{1}{\sqrt{x}} dx, \quad 0 < a < b.$$

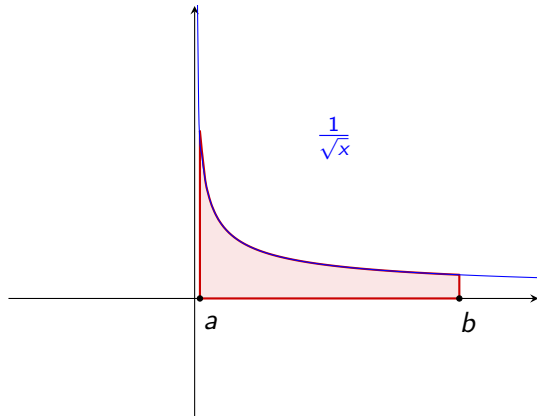
**Fix**  $b$ , as  $a \rightarrow 0^+$ ,

$$\int_a^b \frac{1}{\sqrt{x}} dx \rightarrow L,$$

for some finite  $L$ . But, **fix**  $a$ , as  $b \rightarrow \infty$ ,

$$\int_a^b \frac{1}{\sqrt{x}} dx \rightarrow \infty.$$

<https://www.geogebra.org/m/t5v4p7dz>



# Introduction

To be precise, by the **fundamental theorem of calculus**,

$$\int_a^b \frac{1}{\sqrt{x}} dx = 2 \left( \sqrt{b} - \sqrt{a} \right), \quad 0 < a < b.$$

So, when  $a \rightarrow 0$ ,  $\sqrt{a} \rightarrow 0$ , hence, integral is equal to  $2\sqrt{b}$ . However, as  $b \rightarrow \infty$ ,  $\sqrt{b} \rightarrow \infty$ , and thus the integral is undefined. So, we want to say

$$\int_0^b \frac{1}{\sqrt{x}} dx = 2\sqrt{b}.$$

# Introduction

Consider another example. Let

$$f(x) = \frac{x^2 - 1}{x - 1}, \quad x \neq 1.$$

If  $a < b < 1$ , or  $1 < a < b$ , then by the **fundamental theorem of calculus**,

$$\int_a^b f(x) dx = \int_a^b \frac{(x+1)(x-1)}{(x-1)} dx = \int_a^b x+1 dx = \frac{b^2 - a^2}{2} + b - a.$$

The answer on the right is independent of whether  $a, b = 1$ , we may for example, let  $a = 0$  and  $b = 1$ , then

$$\int_0^1 f(x) dx = \frac{1^2 - 0^2}{2} + 1 - 0 = \frac{3}{2}.$$

But  $f$  has **no antiderivative** over  $[0, 1]$ , in fact,  $f$  is **undefined** at  $x = 1$ .

# Introduction

Consider

$$\int_a^b e^x dx, \quad a < b.$$

By the **fundamental theorem of calculus**,

$$\int_a^b e^x dx = e^b - e^a, \quad a < b.$$

If  $a \rightarrow -\infty$ ,  $e^a \rightarrow 0$ , and thus  $\lim_{a \rightarrow -\infty} \int_a^b e^x dx = e^b$ . Can we say

$$\int_{-\infty}^b e^x dx = e^b?$$

**Improper Integral** allows us to integral a function  $f$  from  $a$  to  $b$ , when either

- ▶  $f$  is **undefined** at some points in  $[a, b]$ ,
- ▶  $f$  is **not continuous** at some points in  $[a, b]$ ,
- ▶ either  $a$  or  $b$  is **not a finite number**.

# Improper Integral

## Definition

Suppose  $f$  is continuous on  $[a, b)$ . The improper integral is defined to be

$$\int_a^b f(x) \, dx = \lim_{t \rightarrow b^-} \int_a^t f(x) \, dx.$$

Suppose  $f$  is continuous on  $(a, b]$ . The improper integral is defined to be

$$\int_a^b f(x) \, dx = \lim_{t \rightarrow a^+} \int_t^b f(x) \, dx.$$

The improper integral  $\int_a^b f(x) \, dx$  is

- ▶ convergent, if the corresponding limits exists;
- ▶ divergent, if the corresponding limits does not exists.

## Example

Evaluate the improper integral

$$\int_2^5 \frac{1}{\sqrt{x-2}} dx.$$

$$\begin{aligned} \int_2^5 \frac{1}{\sqrt{x-2}} dx &= \lim_{t \rightarrow 2^+} \int_t^5 \frac{1}{\sqrt{x-2}} dx = \lim_{t \rightarrow 2^+} 2 \left[ \sqrt{x-2} \right]_t^5 \\ &= \lim_{t \rightarrow 2^+} \left( 2\sqrt{3} - 2\sqrt{t-2} \right) \\ &= 2\sqrt{3}. \end{aligned}$$



# Improper Integral

## Definition

Suppose  $f$  is continuous on  $[a, b] \setminus \{c\}$ . The improper integral is defined to be

$$\int_a^b f(x) \, dx = \int_a^c f(x) \, dx + \int_c^b f(x) \, dx,$$

where  $\int_a^c f(x) \, dx$  and  $\int_c^b f(x) \, dx$  are themselves improper integrals. The integral is

- ▶ convergent if both  $\int_a^c f(x) \, dx$  and  $\int_c^b f(x) \, dx$  are convergent,
- ▶ divergent if either of  $\int_a^c f(x) \, dx$  or  $\int_c^b f(x) \, dx$  is divergent.

## Example

Evaluate the improper integral

$$\int_{-1}^8 \frac{1}{\sqrt[3]{x}} dx.$$

The integrand is continuous on  $[-1, 8] \setminus \{0\}$ .

$$\begin{aligned} \int_{-1}^8 \frac{1}{\sqrt[3]{x}} dx &= \int_{-1}^0 \frac{1}{\sqrt[3]{x}} dx + \int_0^8 \frac{1}{\sqrt[3]{x}} dx = \lim_{t \rightarrow 0^-} \int_{-1}^t \frac{1}{\sqrt[3]{x}} dx + \lim_{t \rightarrow 0^+} \int_t^8 \frac{1}{\sqrt[3]{x}} dx \\ &= \lim_{t \rightarrow 0^-} \left[ \frac{3}{2} x^{\frac{2}{3}} \right]_{-1}^t + \lim_{t \rightarrow 0^+} \left[ \frac{3}{2} x^{\frac{2}{3}} \right]_t^8 = \lim_{t \rightarrow 0^-} \left( \frac{3}{2} (t^{\frac{2}{3}} - 1) \right) + \lim_{t \rightarrow 0^+} \left( \frac{3}{2} (4 - t^{\frac{2}{3}}) \right) \\ &= -\frac{3}{2} + 6 \\ &= \frac{9}{2}. \end{aligned}$$

# Improper Integral

- ▶ Suppose  $f$  is **continuous** on  $[a, \infty)$ . Define the **improper integral** to be

$$\int_a^{\infty} f(x) \, dx = \lim_{b \rightarrow \infty} \int_a^b f(x) \, dx.$$

- ▶ Suppose  $f$  is **continuous** on  $(-\infty, b]$ . Define the **improper integral** to be

$$\int_{-\infty}^b f(x) \, dx = \lim_{a \rightarrow -\infty} \int_a^b f(x) \, dx.$$

- ▶ Suppose  $f$  is **continuous** on  $(-\infty, \infty)$ . Define the **improper integral** to be

$$\int_{-\infty}^{\infty} f(x) \, dx = \int_{-\infty}^c f(x) \, dx + \int_c^{\infty} f(x) \, dx$$

for any real number  $c$ , where  $\int_{-\infty}^c f(x) \, dx$  and  $\int_c^{\infty} f(x) \, dx$  are themselves **improper integrals**.

The **improper integral** is **convergent** if the corresponding limit exists, and **divergent** otherwise.

## Example

For what **rational** values of  $p$  is the integral

$$\int_1^{\infty} \frac{dx}{x^p}$$

convergent?

If  $p \neq 1$ ,

$$\int_1^{\infty} \frac{dx}{x^p} = \lim_{b \rightarrow \infty} \int_1^b \frac{dx}{x^p} = \lim_{b \rightarrow \infty} \left[ \frac{x^{-p+1}}{1-p} \right]_1^b = \lim_{b \rightarrow \infty} \frac{b^{1-p} - 1}{1-p} = \begin{cases} \frac{1}{p-1}, & p > 1 \\ \infty, & p < 1 \end{cases}$$

If  $p = 1$ ,

$$\int_1^{\infty} \frac{dx}{x} = \lim_{b \rightarrow \infty} \int_1^b \frac{dx}{x} = \lim_{b \rightarrow \infty} [\ln(x)]_1^b = \lim_{b \rightarrow \infty} \ln(b) = \infty.$$

In conclusion, the integral converges if and only if  $p > 1$ .

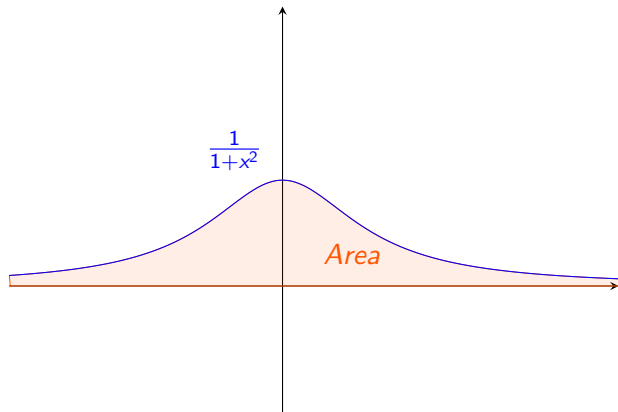
## Example

Find the area bounded by the graph of  $\frac{1}{1+x^2}$  and the x-axis.

That is to compute

$$\int_{-\infty}^{\infty} \frac{dx}{1+x^2}.$$

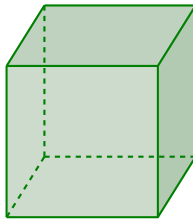
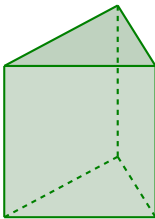
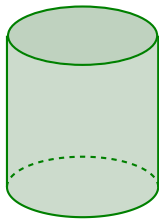
$$\begin{aligned} \int_{-\infty}^{\infty} \frac{dx}{1+x^2} &= \int_{-\infty}^0 \frac{dx}{1+x^2} + \int_0^{\infty} \frac{dx}{1+x^2} \\ &= \lim_{a \rightarrow -\infty} \int_a^0 \frac{dx}{1+x^2} + \lim_{b \rightarrow \infty} \int_0^b \frac{dx}{1+x^2} \\ &= \lim_{a \rightarrow -\infty} [\tan^{-1}(x)]_a^0 + \lim_{b \rightarrow \infty} [\tan^{-1}(x)]_0^b \\ &= \lim_{a \rightarrow -\infty} -\tan^{-1}(a) + \lim_{b \rightarrow \infty} \tan^{-1}(b) \\ &= -\left(-\frac{\pi}{2}\right) + \frac{\pi}{2} = \pi. \end{aligned}$$



## Volume using Cross-Sections

## Method of Slicing

Consider the following cylindrical objects that is defined by a 2-dimensional surface.



The **volume** of such objects are given by  $A \times h$ , where

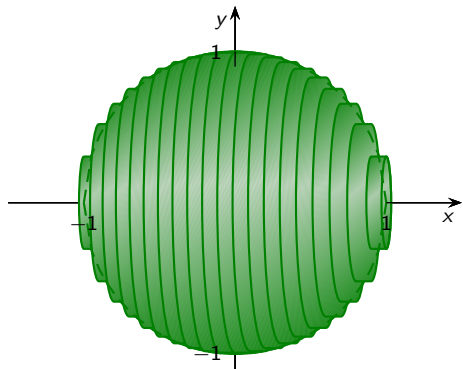
- ▶  $A$  is the surface area of the 2-dimensional surface defining the object, and
- ▶  $h$  is the height of the object.

### Example

- ▶ Volume of **Cylinder**  $= \pi r^2 \times h$
- ▶ Volume of a **Cuboid**  $= lb \times h$

# Method of Slicing

Consider now the **unit sphere**. Place it in on  $-1 \leq x \leq 1$ .



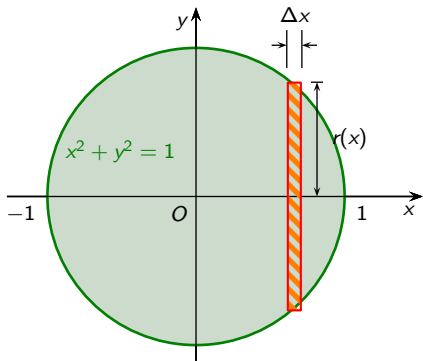
May approximate the **unit sphere** with **cylinders**.

- ▶ Partition the interval  $[-1, 1]$  into  $n$  equal width partitions,  $P = \{x_0 = -1, x_1 = -1 + \frac{2}{n}, \dots, x_k = -1 + \frac{2k}{n}, \dots, x_n = 1\}$ .
- ▶ The base area of the  $k$ -th cylinder is given by some function  $A(c_k)$ .
- ▶ The height of each cylinder is the width of the partition,  $\Delta x = \frac{2}{n}$ .



## Method of Slicing

The function  $A(x)$  is given as such.



The volume of the sphere is hence

$$\lim_{n \rightarrow \infty} \sum_{k=1}^n A(x_k) \Delta x = \int_{-1}^1 \pi(1 - x^2) dx = \pi \left[ x - \frac{x^3}{3} \right]_{-1}^1 = \frac{4}{3}\pi.$$

- ▶ The radius of the cylinder is  $r(x) = \sqrt{1 - x^2}$ .
- ▶ Thus the base area is  $A(x) = \pi r(x)^2 = \pi(1 - x^2)$
- ▶ The volume of the cylinder is therefore  $\pi(1 - x^2)\Delta x$ .

# Method of Slicing

## Definition

The **volume** of a solid of **integrable cross-sectional area**  $A(x)$  from  $x = a$  to  $x = b$  is the integral of  $A$  from  $a$  to  $b$ ,

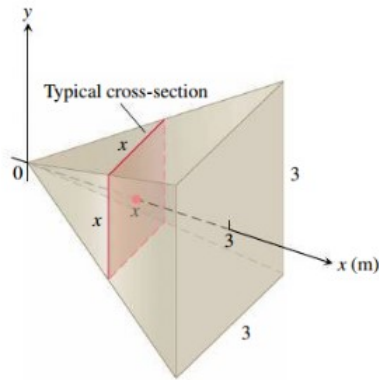
$$V = \int_a^b A(x) \, dx.$$

## Algorithm.

1. Sketch the **solid** and a typical **cross-section**.
2. Find a **formula** for  $A(x)$ , the area of a typical **cross-section**.
3. Find the **limits of integration**.
4. **Integrate**  $A(x)$  to find the **volume**.

## Example

A pyramid 3m high has a square base that is 3m on a side. The cross-section of the pyramid perpendicular to the altitude  $x$  m down from the vertex is a square  $x$  m on a side. Find the volume of the pyramid.



The typical cross-section is a square, with sides of length  $x$  m. Hence,

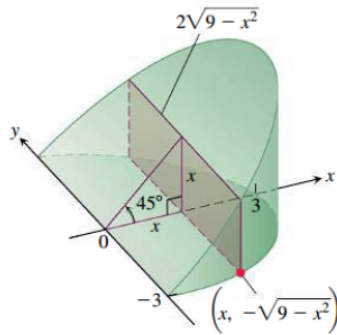
$$A(x) = x^2.$$

The limits of integration are  $x = 0$  to  $x = 3$ . Therefore, the volume is

$$V = \int_0^3 x^2 dx = \left[ \frac{x^3}{3} \right]_0^3 = 9m^3.$$

## Example

A curved wedge is cut from a circular cylinder of radius 3 by two planes. One plane is perpendicular to the axis of the cylinder. The other plane crosses the first plane at  $45^\circ$  angle



The typical cross-section is a rectangle, with dimensions

$$(2\sqrt{9-x^2}) \times x \Rightarrow A(x) = 2x\sqrt{9-x^2}.$$

The limits of the integral are  $x=0$  and  $x=3$ . Hence, the volume is

$$V = \int_0^3 2x\sqrt{9-x^2} dx = - \int_9^0 \sqrt{u} du = \frac{2}{3} \left[ u^{\frac{3}{2}} \right]_0^9 = 18,$$

where we used the substitution  $u(x) = 9 - x^2$ .

# Solid of Revolution: The Disk Method

Suppose  $f(x)$  is a continuous function. A solid is formed by rotating the region bounded between  $f(x)$  and the  $x$ -axis about the  $x$ -axis.

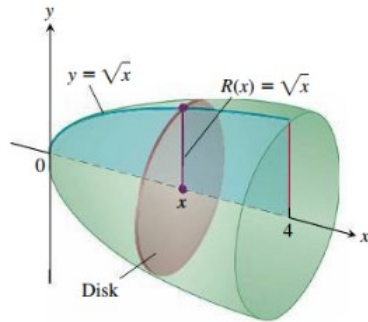
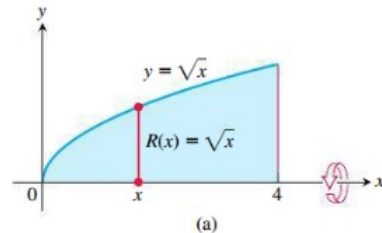
Now take each slice of the solid perpendicular to the  $x$ -axis, it is a **disk** with radius  $|f(x)|$ , and thus, the cross-sectional area is

$$A(x) = \pi |f(x)|^2 = \pi f(x)^2.$$

This method for calculating the volume of a solid of revolution is called the **disk method**.

## Example

The volume of the sphere computed above is the disk method.



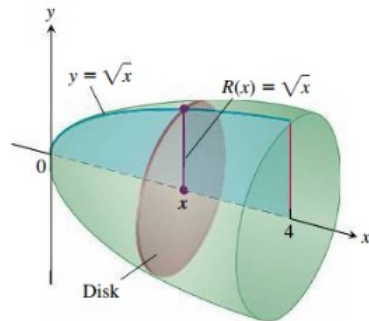
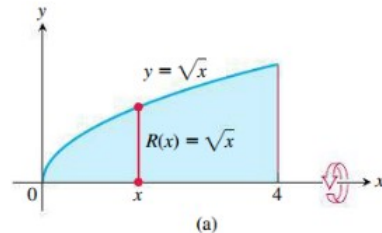
## Example

The function is  $f(x) = \sqrt{x}$ . So, the cross-sectional area is

$$A(x) = \pi x.$$

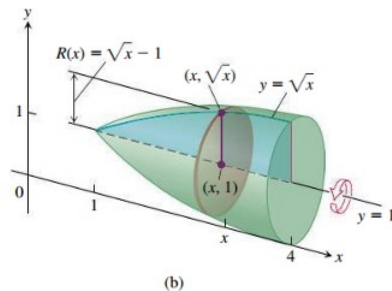
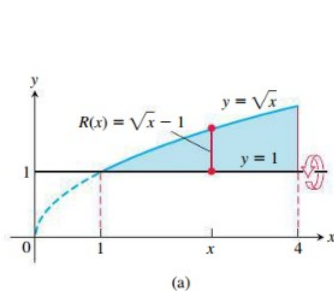
The limits of integration are  $x = 0$ ,  $x = 4$ . So, the volume is

$$V = \int_0^4 \pi x dx = \pi \left[ \frac{x^2}{2} \right]_0^4 = 8\pi.$$



## Example

Find the volume of the solid generated by revolving the region bounded by  $f(x) = \sqrt{x}$  and the lines  $y = 1$ ,  $x = 4$ , about the line  $y = 1$ .



The radius of each disk is now  $\sqrt{x} - 1$ , and so the cross-sectional area is  $A(x) = \pi (\sqrt{x} - 1)^2$ . The limits of integration are  $x = 1$  and  $x = 4$ . Hence, the volume is

$$V = \int_1^4 \pi (\sqrt{x} - 1)^2 dx = \pi \int_1^4 x - 2\sqrt{x} + 1 dx = \pi \left[ \frac{x^2}{2} - 2 \cdot \frac{2}{3} x^{\frac{3}{2}} + x \right]_1^4 = \frac{7\pi}{6}.$$

## Example

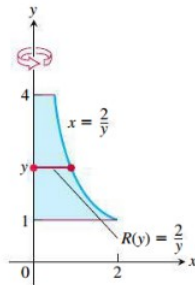
Find the volume of the solid generated by revolving the region between the  $y$ -axis and the curve  $x = \frac{2}{y}$ ,  $1 \leq y \leq 4$ , about the  $y$ -axis.

Function of  $y$ ,  $f(y) = \frac{2}{y}$ . The area of the disk is

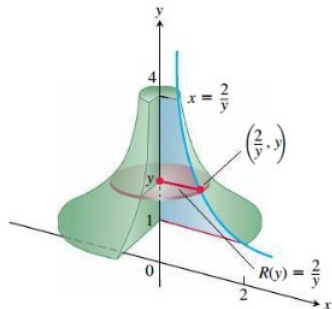
$$A(y) = \pi \left( \frac{2}{y} \right)^2 = \frac{4\pi}{y^2}.$$

The limits of integration are  $y = 1$  to  $y = 4$ . Therefore, the volume is

$$V = \int_1^4 \frac{4\pi}{y^2} dy = -4\pi [y^{-1}]_1^4 = 3\pi.$$



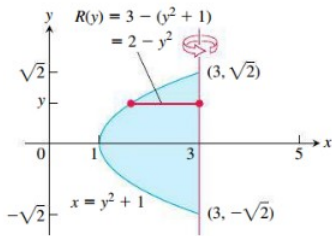
(a)



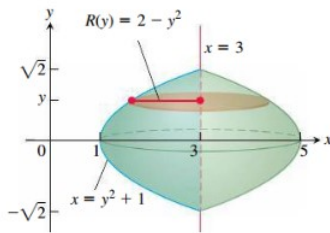


## Example

Find the volume of the solid generated by revolving the region between the parabola  $x = y^2 + 1$  and the line  $x = 3$  about the line  $x = 3$ .



(a)



(b)

The radius is  $3 - (y^2 + 1) = 2 - y^2$ , and so, the cross-sectional area is  $A(x) = \pi (2 - y^2)^2$ . To find the limits of integration, solve  $3 = y^2 + 1 \Rightarrow$  the limits are  $y = -\sqrt{2}$ ,  $y = \sqrt{2}$ . Hence, the volume is

$$V = \int_{-\sqrt{2}}^{\sqrt{2}} \pi (2 - y^2)^2 dy = \pi \int_{-\sqrt{2}}^{\sqrt{2}} 4 - 4y^2 + y^4 dy = \pi \left[ 4y - \frac{4y^3}{3} + \frac{y^5}{5} \right]_{-\sqrt{2}}^{\sqrt{2}} = \frac{64\pi\sqrt{2}}{15}.$$

## Solids of Revolution: The Washer Method

Suppose now the region we revolve does not border on or cross the axis of revolution, but is bounded by 2 functions. Then the cross-sections perpendicular to the axis of revolution are **washers** instead of disks. The dimension of a typical washer are

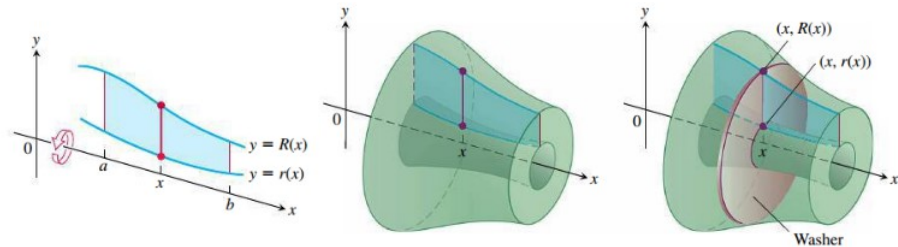
Outer Radius :  $R(x)$

Inner Radius :  $r(x)$

Then the cross-sectional area is

$$A(x) = \pi (R(x)^2 - r(x)^2).$$

This method for calculating the volume of a solid of revolution is called the **washers method**.

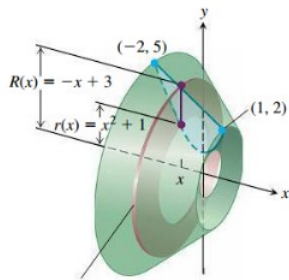
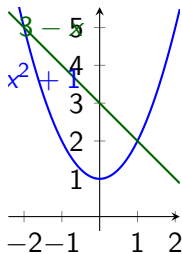


## Example

The region bounded by the curve  $y = x^2 + 1$  and the line  $y = -x + 3$  is revolved about the  $x$ -axis to generate a solid. Find the volume of the solid.

The intersection between  $x^2 + 1$  and  $-x + 3$  are  $x = -2$  and  $x = 1$ . In this region,  $3 - x \geq x^2 + 1$ . So,  $R(x) = 3 - x$ ,  $r(x) = x^2 + 1$ . Hence, the volume is

$$\begin{aligned} V &= \int_{-2}^1 \pi ((3-x)^2 - (x^2+1)^2) dx = \pi \int_{-2}^1 9 - 6x + x^2 - x^4 - 2x^2 - 1 dx \\ &= \int_{-2}^1 8 - 6x - x^2 - x^4 dx = \left[ 8x - 3x^2 - \frac{x^3}{3} - \frac{x^5}{5} \right]_{-2}^1 \\ &= \frac{117\pi}{5} \end{aligned}$$



Washer cross-section

Outer radius:  $R(x) = -x + 3$

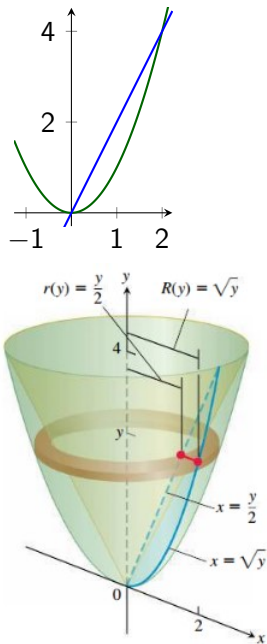
Inner radius:  $r(x) = x^2 + 1$

## Example

The region bounded by the parabola  $y = x^2$  and the line  $y = 2x$  in the first quadrant is revolved about the  $y$ -axis to generate a solid. Find the volume of the solid.

Since we are revolving about the  $y$ -axis, we need the functions in terms of  $y$ . The intersection between  $\sqrt{y}$  and  $\frac{y}{2}$  is  $y = 0$  and  $y = 4$ . In this region,  $\sqrt{y} > \frac{y}{2}$ . So,  $R(y) = \sqrt{y}$ ,  $r(y) = \frac{y}{2}$ . The volume is

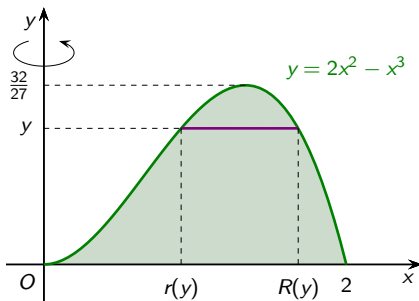
$$V = \int_0^4 \pi \left( y - \frac{y^2}{4} \right) dy = \pi \left[ \frac{y^2}{2} - \frac{y^3}{12} \right]_0^4 = \frac{8\pi}{3}.$$



## Volumes using Cylindrical Shells

## Introduction

Find the volume of the solid formed by rotating the region bounded by  $y = 2x^2 - x^3$  and  $y = 0$  about the  $y$ -axis.



## Problem

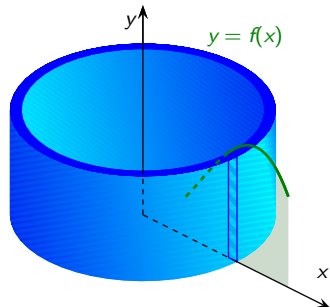
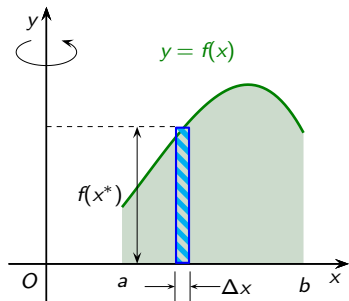
Cannot write the boundary as a function of  $y$ .

Cannot split the boundary into 2 functions  $R(y)$  and  $r(y)$ .

**Conclusion:** Cannot use washer method.

## The Shell Method

Instead, we partition  $[a, b]$  into  $P = \{x_0 = a, \dots, x_n = b\}$  we consider a cylinder of thickness  $\Delta x_k$ , height  $f(c_k)$ , and have radius  $r(c_k)$ . Then the volume of the cylinder is  $V(x_k) = 2\pi r(c_k)f(c_k)\Delta x$ .

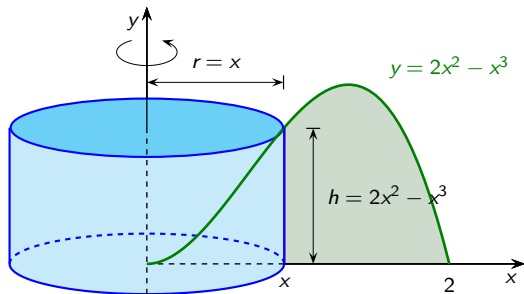


Hence, the volume of the solid is

$$\lim_{n \rightarrow \infty} \sum_{k=1}^n V(k) = \lim_{n \rightarrow \infty} \sum_{k=1}^n 2\pi r(c_k) f(c_k) \Delta x = \int_a^b 2\pi r(x) f(x) dx.$$

## Example

Find the volume of the solid formed by rotating the region bounded by  $y = 2x^2 - x^3$  and  $y = 0$  about the  $y$ -axis.



Using the shell method, the height of the cylinder is  $f(x) = 2x^2 - x^3$ , and the radius is  $r(x) = x$ . Hence, the volume is

$$\int_0^2 2\pi (2x^3 - x^4) dx = 2\pi \left[ \frac{x^4}{2} - \frac{x^5}{5} \right]_0^2 = \frac{16\pi}{5}.$$



# The Shell Method

## Definition (Shell formula for revolution about a vertical line)

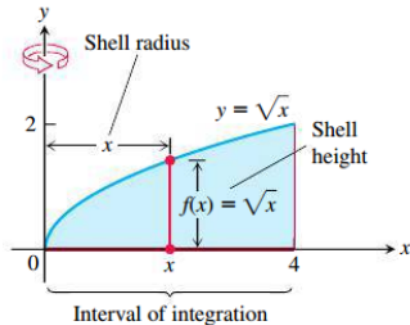
The volume of the solid generated by revolving the region between the  $x$ -axis and the graph of a continuous function  $f(x) \geq 0$ ,  $L \leq a \leq x \leq b$ , about a vertical line  $x = L$  is

$$V = \int_a^b 2\pi r(x) f(x) dx,$$

where  $r(x) = x - L$  is the **radius** of the shell. This method for calculating the volume of a solid of revolution is called the **shell method**.

## Example

The region bounded by the curve  $y = \sqrt{x}$ , the  $x$ -axis, and the line  $x = 4$  is revolved about the  $y$ -axis to generate a solid. Find the volume of the solid.

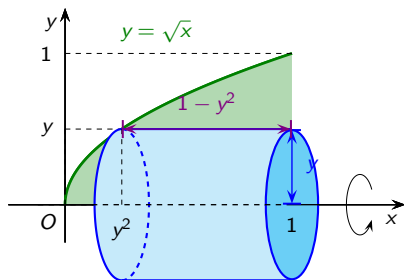


The shell radius is  $r(x) = x$ , shell height is  $f(x) = \sqrt{x}$ . Therefore, the volume is

$$V = \int_0^4 2\pi x \sqrt{x} dx = 2\pi \left[ \frac{2}{5} x^{\frac{5}{2}} \right]_0^4 = \frac{128\pi}{5}$$

## Example

The region bounded by the curve  $y = \sqrt{x}$ , the  $y$ -axis, and the line  $x = 1$  is revolved about the  $x$ -axis to generate a solid. Find the volume of the solid using the shell method.

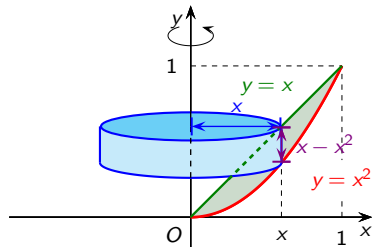


Making  $y$  the subject of the fomular, we have  $x = y^2$ . The shell radius is  $r(y) = y$  and the shell height is  $h(y) = 1 - y^2$ . Hence, the volume is

$$V = \int_0^1 2\pi y(1 - y^2) dy = 2\pi \left[ \frac{y^2}{2} - \frac{y^4}{4} \right]_0^1 = \frac{\pi}{2}.$$

## Example

The region bounded by the curves  $y = x$  and  $y = x^2$  is revolved about the  $y$ -axis to generate a solid. Find the volume of the solid using the shell method.



The shell radius is  $r(x) = x$ , and the shell height is  $h(x) = x - x^2$ . The intersection between the curves are at  $x = 0$  and  $x = 1$ . So, the volume is

$$V = \int_0^1 2\pi x (x - x^2) dx = 2\pi \left[ \frac{x^3}{3} - \frac{x^4}{4} \right]_0^1 = \frac{\pi}{6}.$$

**Exercise:** Compute the volume using the washer method.